Biogeography with Macroevolutionary Models: Faster likelihood calculations for State-dependent Speciation-Extinction (SSE) models using *Julia*

Nicholas J. Matzke
Senior Lecturer,
Rutherford Discovery Fellow
School of Biological Sciences
University of Auckland

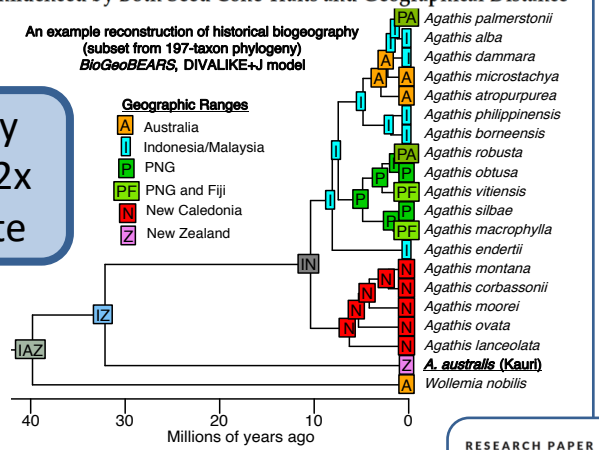
Historical biogeography with new models

Example papers:

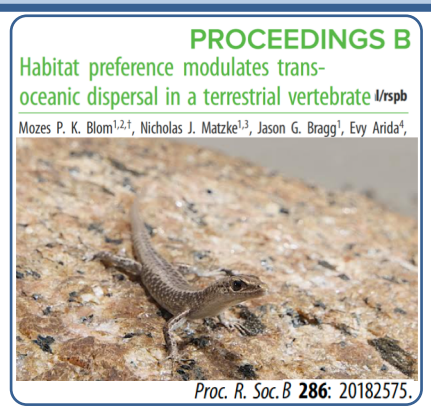
2020: Fleshy cones have 2x dispersal rate

KRISTINA V. KLAUS[1,*] AND NICHOLAS J. MATZKE[2,3]

Statistical Comparison of Trait-Dependent Biogeographical Models Indicates That Podocarpaceae Dispersal Is Influenced by Both Seed Cone Traits and Geographical Distance



2019: *Cryptoblepharus* lizards, littoral ~13x more dispersal



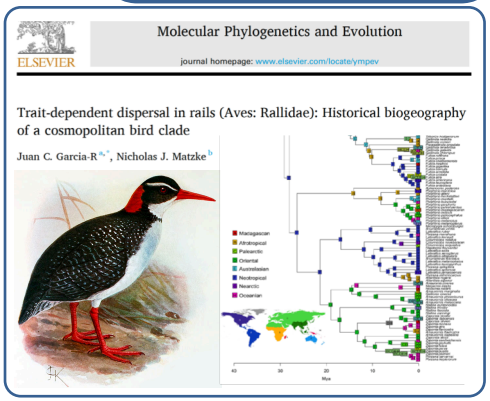
PROCEEDINGS B

Habitat preference modulates trans-oceanic dispersal in a terrestrial vertebrate

Mozes P. K. Blom[1,2,†], Nicholas J. Matzke[1,3], Jason G. Bragg[1], Evy Arida[4],

Proc. R. Soc. B 286: 20182575.

2021: Polyploidy in NZ plants; flightlessness in Pacific rails



ELSEVIER

Molecular Phylogenetics and Evolution

journal homepage: www.elsevier.com/locate/ympev

Trait-dependent dispersal in rails (Aves: Rallidae): Historical biogeography of a cosmopolitan bird clade

Juan C. Garcia-R[a,*], Nicholas J. Matzke[b]

2019: Dispersal syndrome in crocs

RESEARCH PAPER

Global Ecology and Biogeography

A Journal of Macroecology

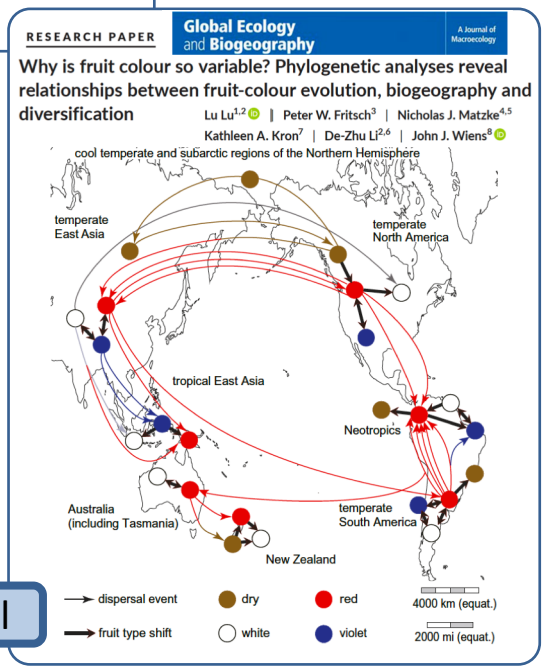
WILEY

Trait-based range expansion aided in the global radiation of Crocodylidae

Michaël P. J. Nicolai[1,2] | Nicholas J. Matzke[3,4]

SEE THE AMAZING VIDEO ONLINE AT Dinosaurnews.com.au

2019: Red fruits ~20x more dispersal



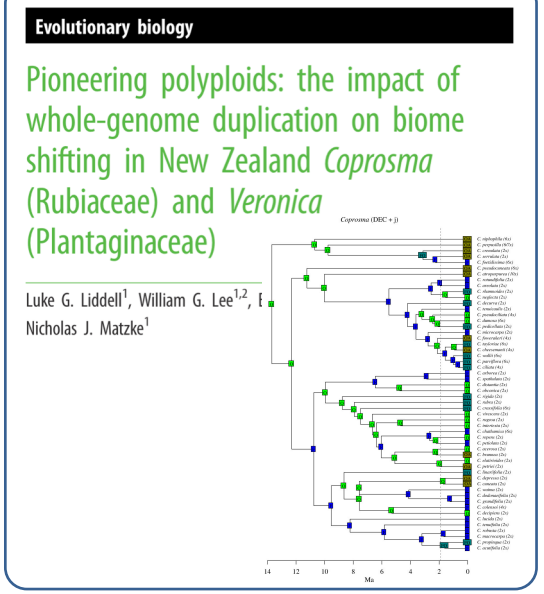
RESEARCH PAPER

Global Ecology and Biogeography

A Journal of Macroecology

Why is fruit colour so variable? Phylogenetic analyses reveal relationships between fruit-colour evolution, biogeography and diversification

Lu Lu[1,2] | Peter W. Fritsch[3] | Nicholas J. Matzke[4,5]
Kathleen A. Kron[7] | De-Zhu Li[2,6] | John J. Wiens[8]



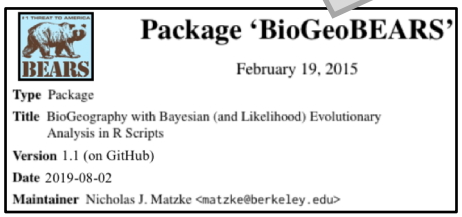
Evolutionary biology

Pioneering polyploids: the impact of whole-genome duplication on biome shifting in New Zealand *Coprosma* (Rubiaceae) and *Veronica* (Plantaginaceae)

Luke G. Liddell[1], William G. Lee[1,2], E
Nicholas J. Matzke[1]

Matzke key product: Biogeographical software

2019: 450+ usages



Package 'BioGeoBEARS'

February 19, 2015

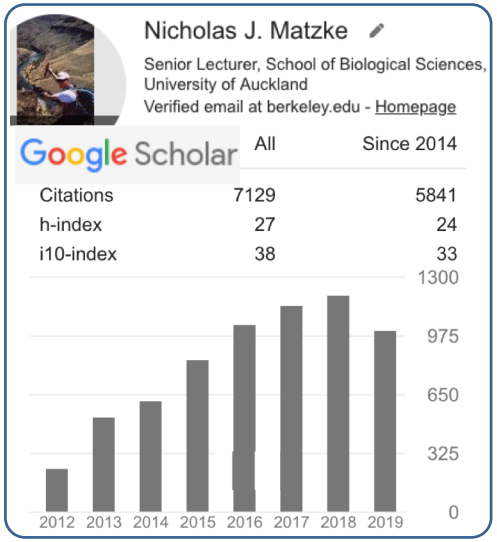
Type Package

Title BioGeography with Bayesian (and Likelihood) Evolutionary Analysis in R Scripts

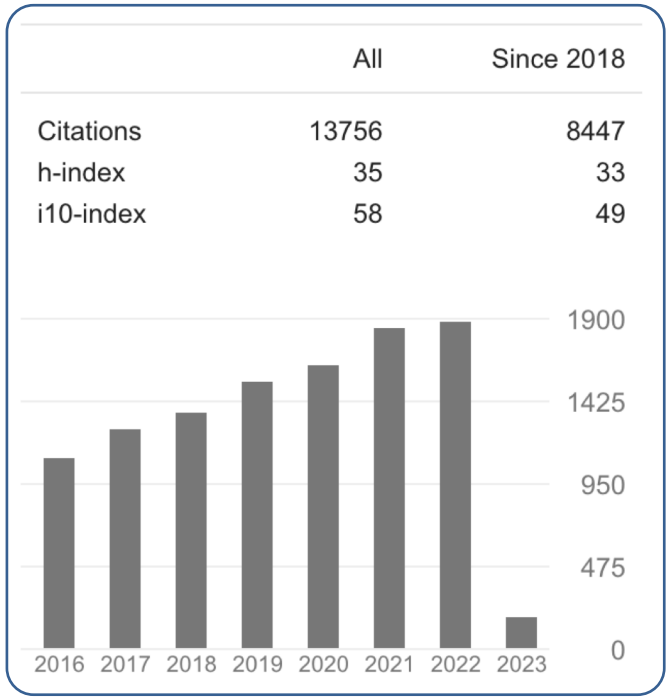
Version 1.1 (on GitHub)

Date 2019-08-02

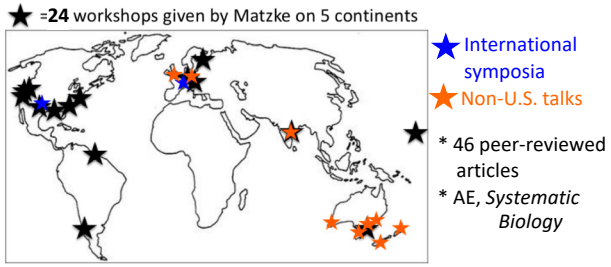
Maintainer Nicholas J. Matzke <matzke@berkeley.edu>



2023: ~1800 usages



Biogeography Workshops:



Controversy! Are these models any good?

A critique of the *BioGeoBEARS* model-comparison approach was published...

DOI: 10.1111/jbi.13173

PERSPECTIVE

WILEY Journal of Biogeography

Conceptual and statistical problems with the DEC+J model of founder-event speciation and its comparison with DEC via model selection

My 2021 rebuttal in the *Journal of Biogeography*:

DOI: 10.1111/jbi.14346

SYNTHESIS

Journal of Biogeography WILEY

Statistical comparison of DEC and DEC+J is identical to comparison of two ClaSSE submodels, and is therefore valid

Nicholas J. Matzke

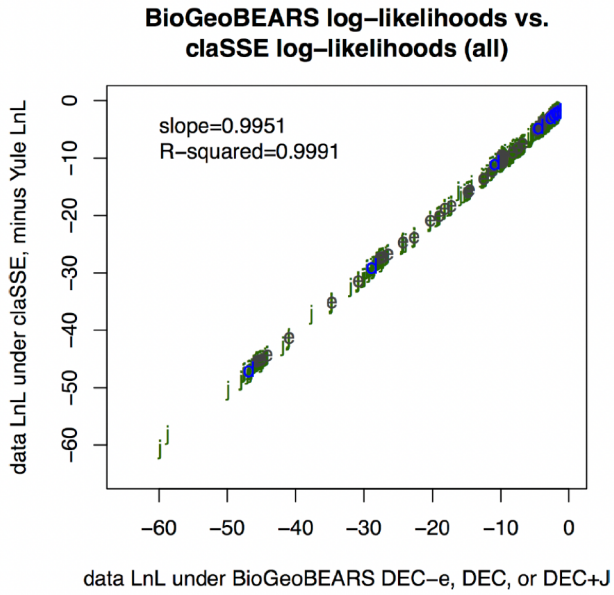


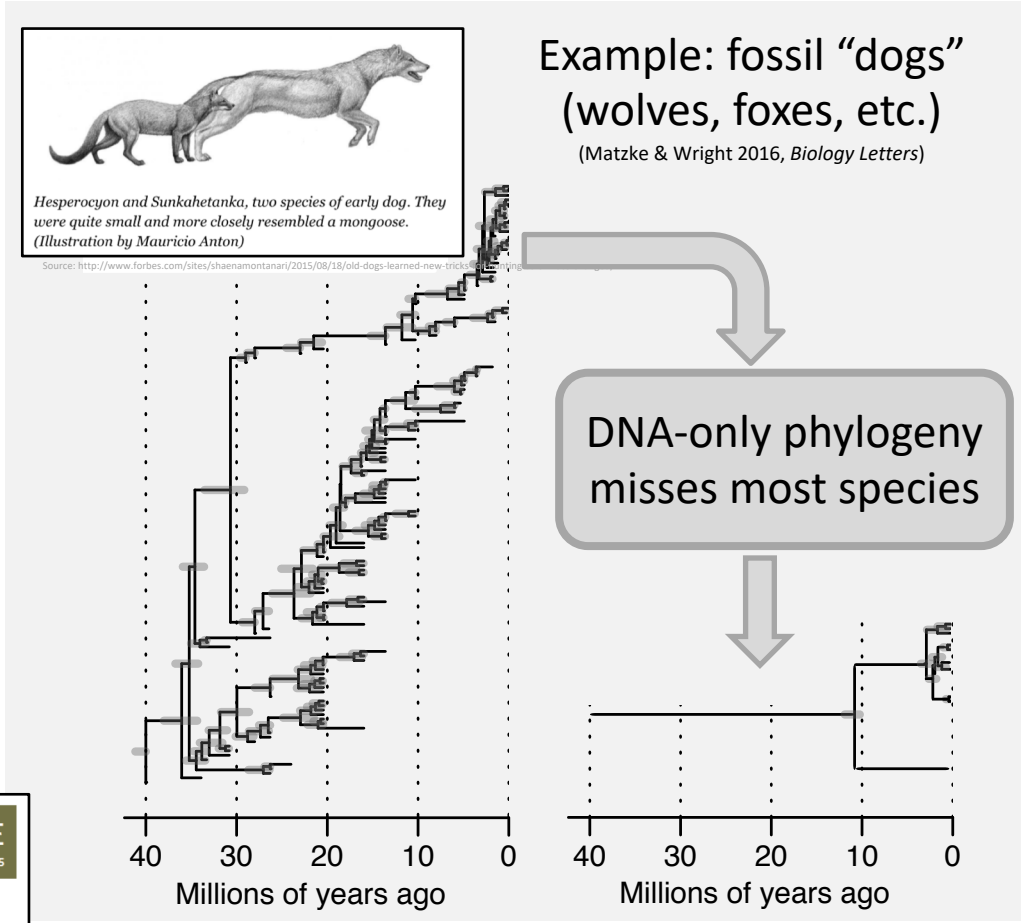
FIGURE 5 Log-likelihoods calculated under BioGeoBEARS DEC and DEC+J for a wide range of parameter values, regressed against log-likelihoods calculated under diversitree ClaSSE models set up to produce DEC and DEC+J behaviour (as in the Figure). The letter *d* represents the DEC model when $e = 0, j = 0$. The letter *e* represents the DEC model when $d > 0, e > 0, j = 0$. The letter *j* represents the DEC+J model when $j > 0$. DEC, Dispersal–Extinction–Cladogenesis

The real challenge to our models: extinction

There are <u>3 HUGE problems</u> with routine biogeographic models:

1. **Extinction is missing**
   (and diversity-dependence)
2. **Fossils not used**
3. **Palaeogeography not used**

"[M]olecular phylogenetic analyses that yield **zero or low extinction…are almost certainly wrong** […] they fail to take into account the origin of extinct species."
(Charles Marshall, 2017, *Nature Ecology & Evolution*)



Summary of key issues:

1. Most biogeographical inference methods *ignore extinction*. This is a big problem

2. There are models (SSE: State-dependent Speciation/Extinction models) that take into account extinction, but they require large systems of Ordinary Differential Equations (ODEs) which are computationally challenging.

3. Part of modern evolutionary biology, and evolutionary biogeography, is thus highly computational: designing solvers that will allow realistic inference to occur fast enough for biologists to use routinely for their research studies.

4. This Marsden Project solved these computational problems with the *Julia* package, *PhyBEARS.jl*, available at: https://github.com/nmatzke/PhyBEARS.jl

Computational Research

Motivation

Some evolutionary models naturally have large state spaces

E.g., biogeography…

Trait-dependent dispersal in Podocarpaceae+Araucariaceae (with Kristina Klaus, Univ. of Bochum)

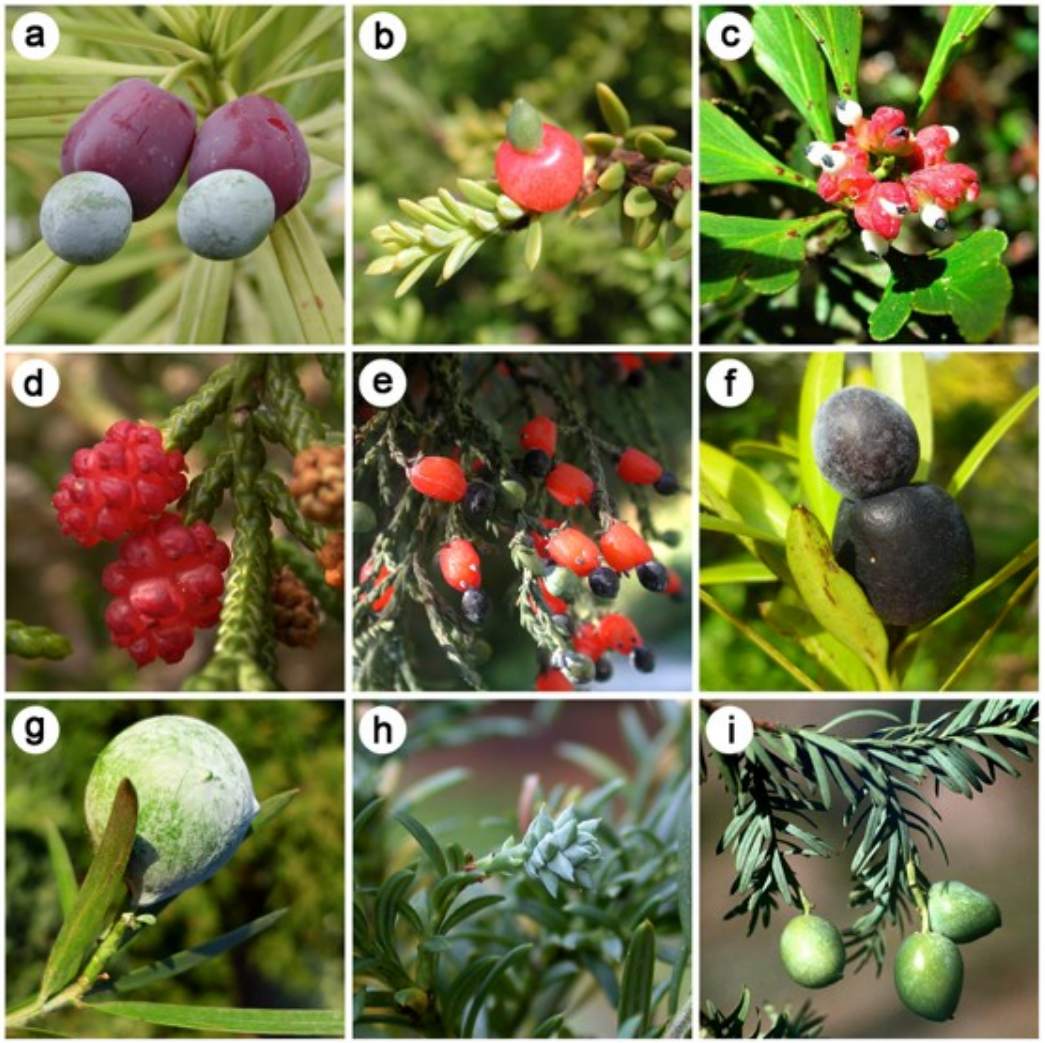


Figure 1. Seed cone variability in the Podocarpaceae. **a** *Podocarpus macrophyllus*, **b** *Podocarpus nivalis*, **c** *Phyllocladus asplenifolius*, **d** *Microcachys tetragona*, **e** *Dacrycarpus dacrydioides* **f** *Podocarpus elatus*, **g** *Afrocarpus falcatus*, **h** *Saxegothaea conspicua*, **i** *Prumnopitys amara*.

Kristina Klaus, Nick Matzke, *Systematic Biology*, 2020

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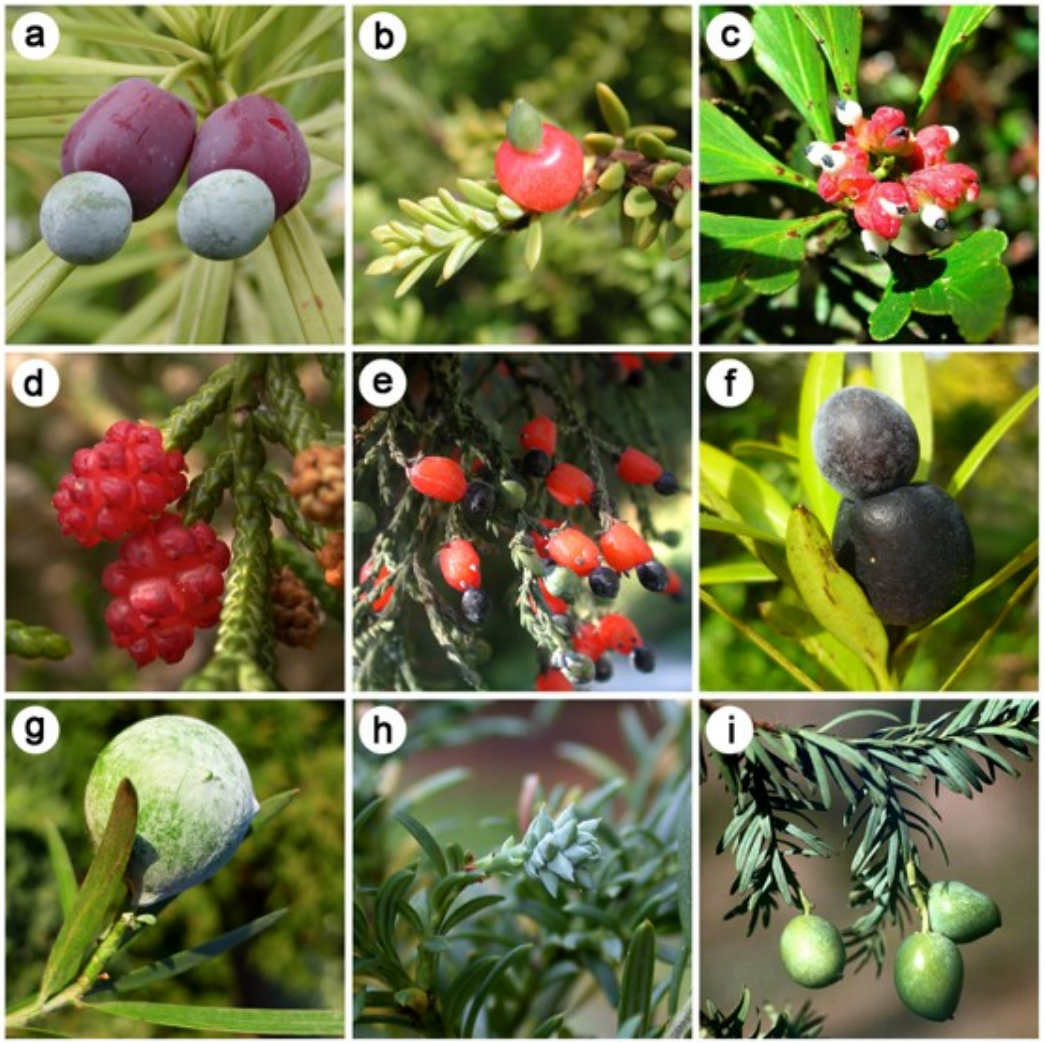

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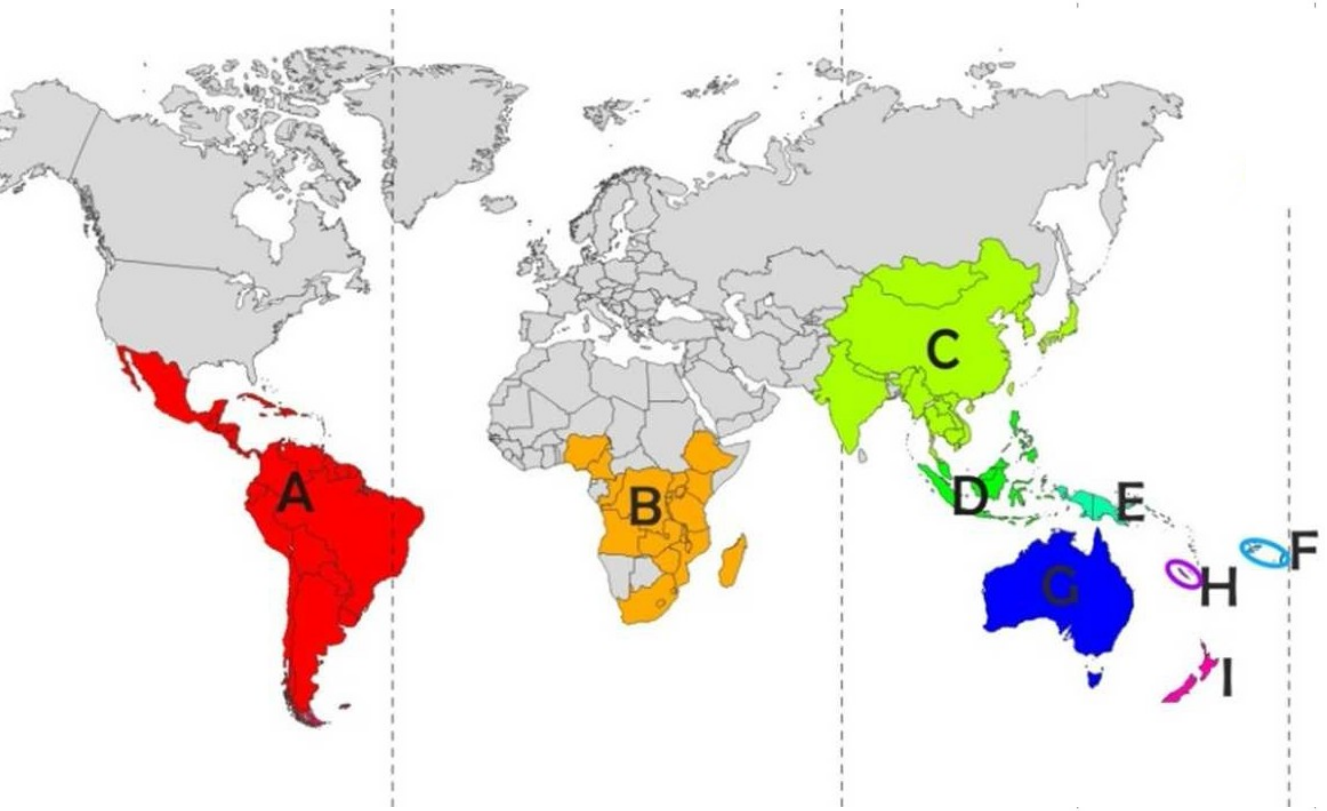
Hypothesis:

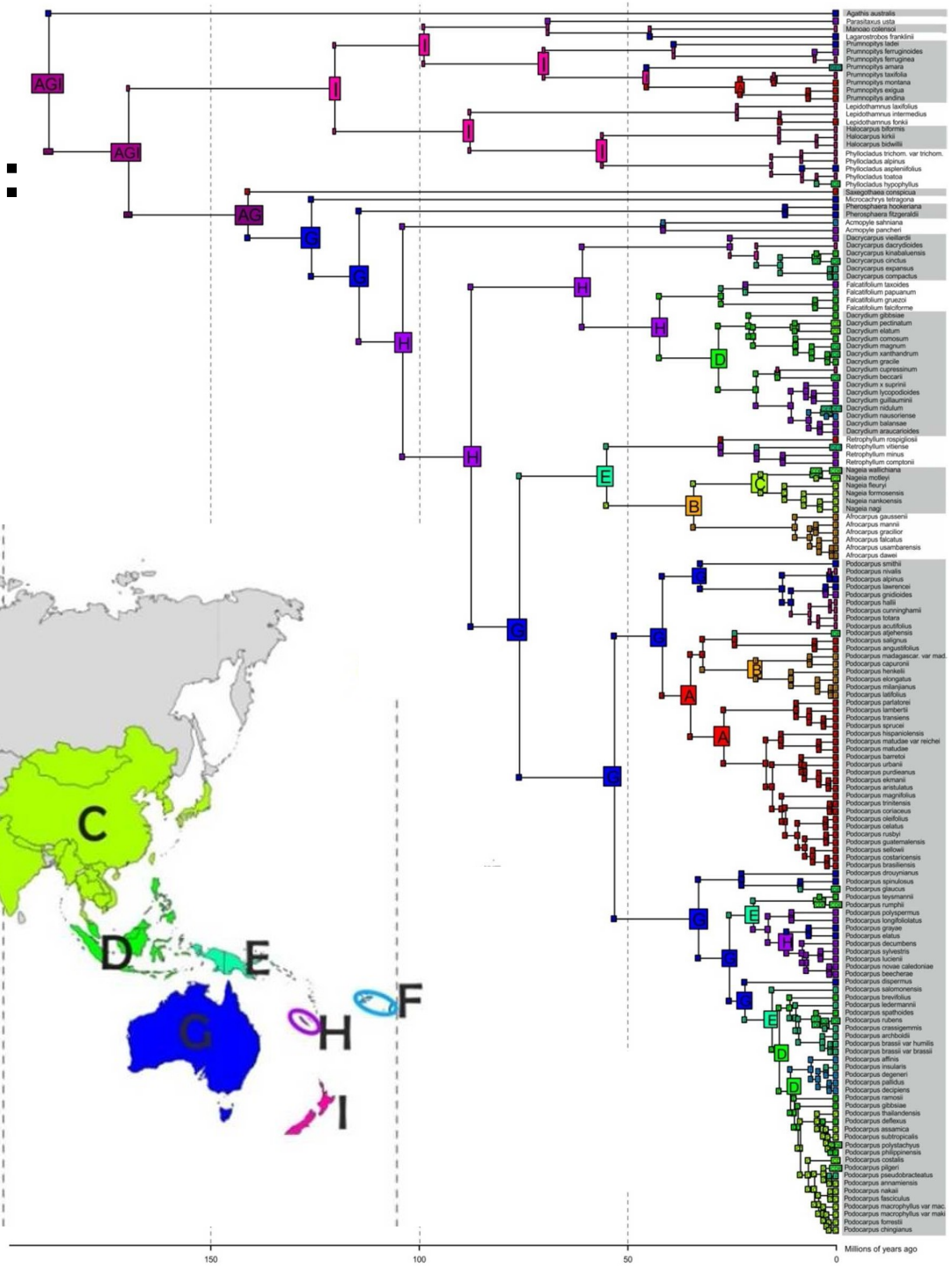
fleshy cones will be better dispersers

Kristina Klaus, Nick Matzke, *Systematic Biology*, 2020

Podocarpaceae DEC+*j*+*x*+traits model:

Biogeography





Kristina Klaus, Nick Matzke, *Systematic Biology*, 2020

Podocarpaceae DEC+*j*+*x*+traits model:

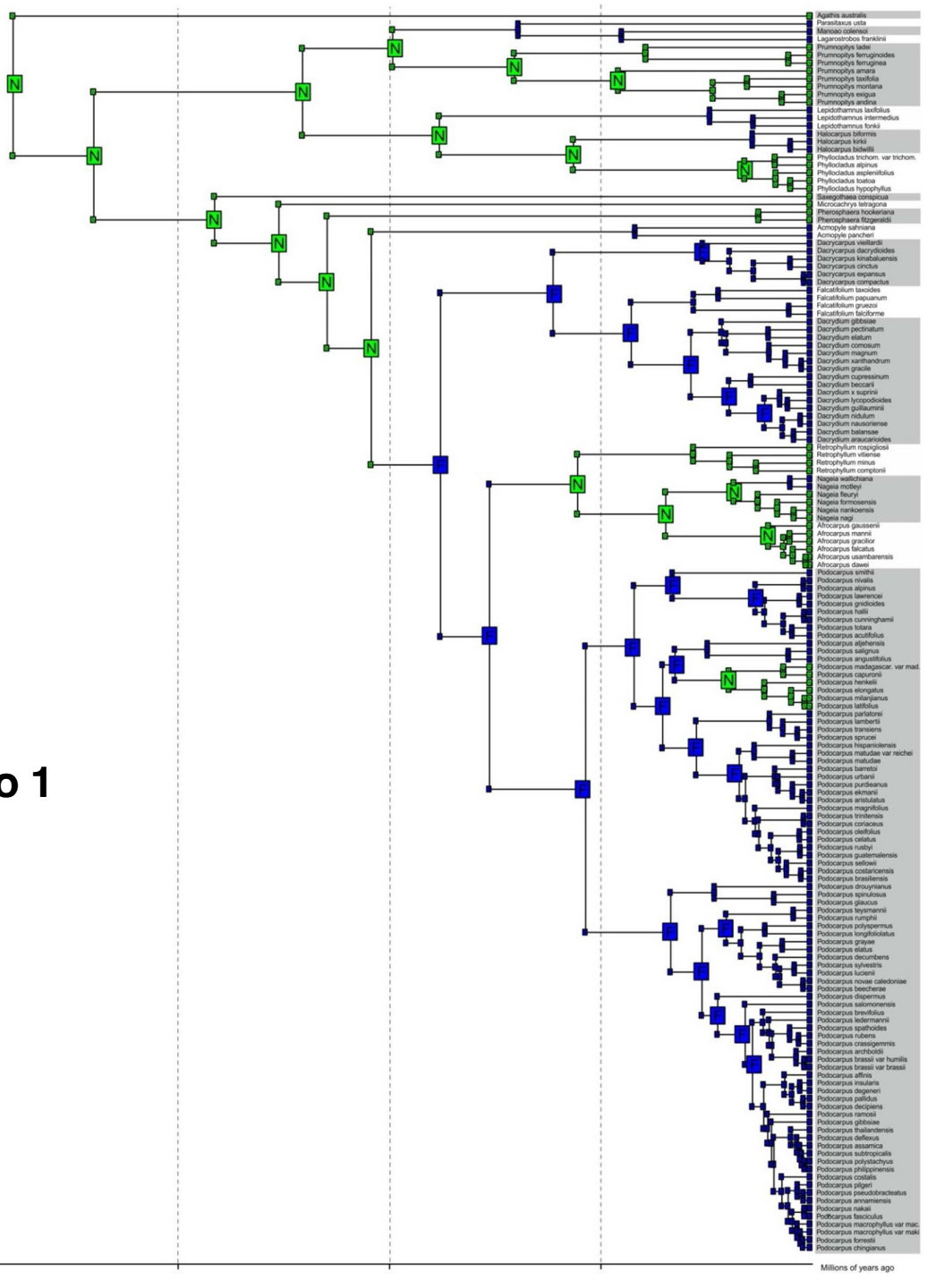
Traits

Fleshy cones

- Dispersal multiplier m_1 fixed to 1

Non-fleshy cones

- Dispersal multiplier $m_2 = 0.5$



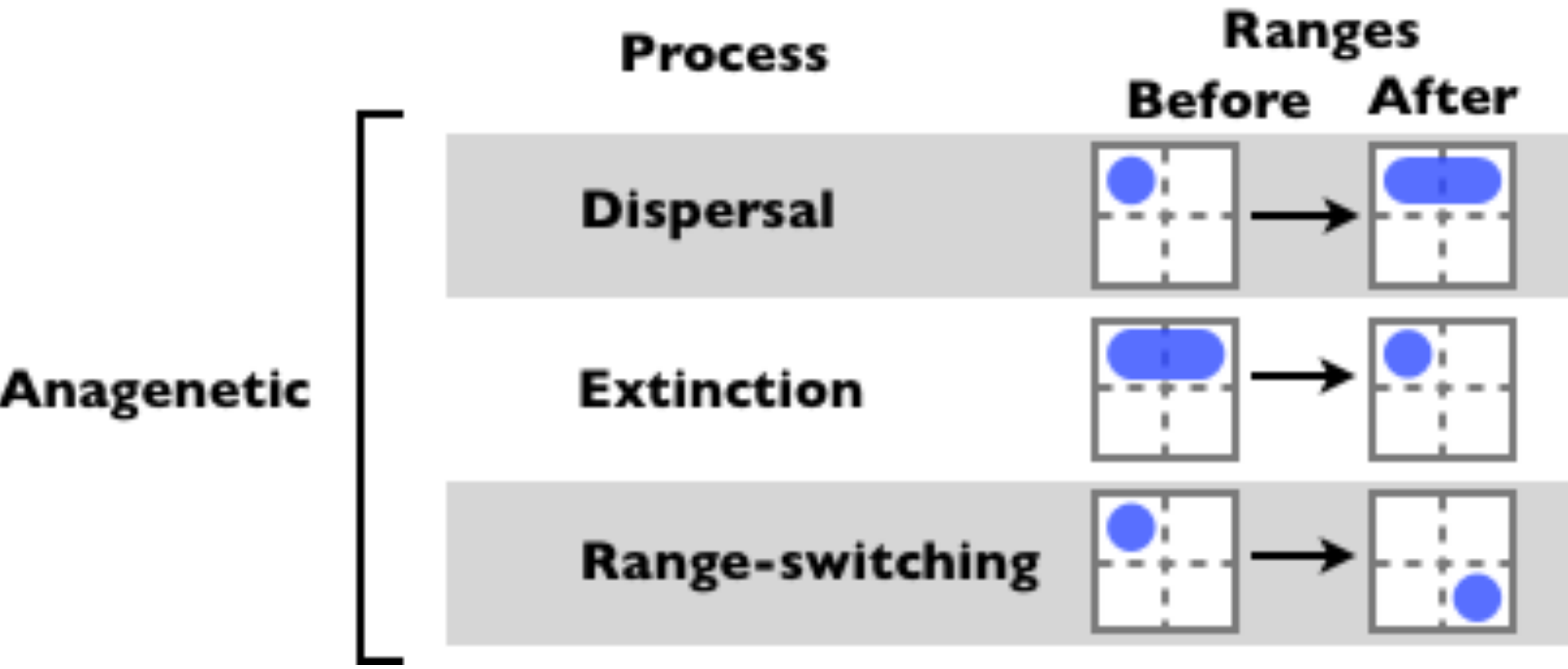
Kristina Klaus, Nick Matzke, *Systematic Biology*, 2020

Geography matrix

descendant range

ancestor range	*null*	A	B	AB
null	-			
A	*e*	-		*d*
B	*e*		-	*d*
AB		*e*	*e*	-

$= Q$


Process
Ranges
Before
After
Anagenetic
Dispersal
Extinction
Range-switching

Problem:

of ranges=$2^{(\#\ \text{of areas})}$

2 areas = 4 states

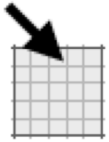
4 areas = 16 states

10 areas = 1024 states

11 areas = 2048 states

DEC model, 12 areas = 2^12 = 4096 states

4x4 rate matrix



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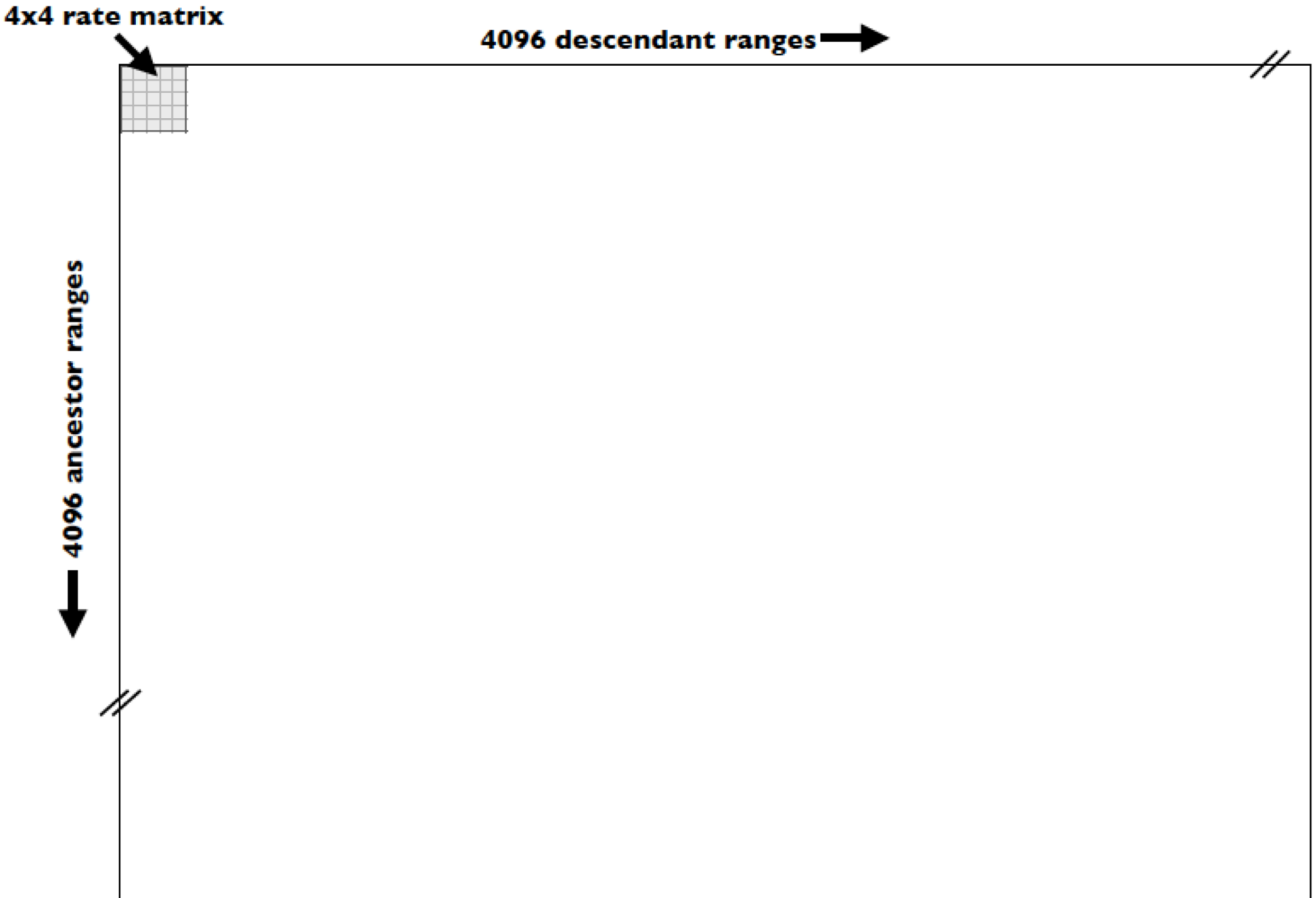

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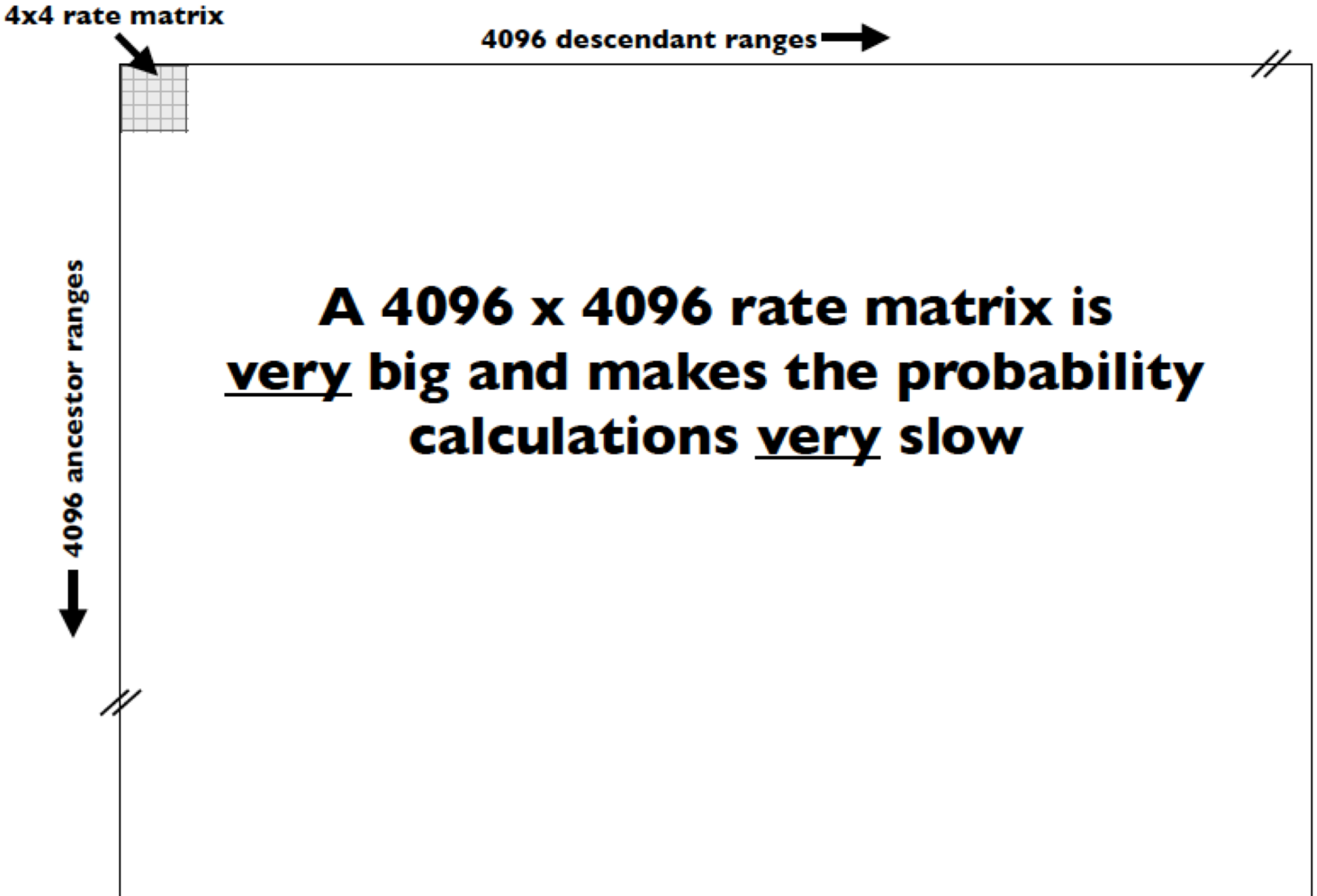

Faster likelihood calculations for State-dependent Speciation-Extinction (SSE) models using Julia

Nicholas J. Matzke, Senior Lecturer, SBS, U. Auckland

Motivation

1. Some evolutionary models naturally have large state spaces

   E.g., biogeography…

2. It gets worse when you add traits:
   …every binary trait doubles the state space

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1. Some evolutionary models naturally have large state spaces

   E.g., biogeography…

2. It gets worse when you add traits

3. Even these slow models ignore extinction (!!)

State-dependent speciation-extinction (SSE) models include extinction…

We have to integrate 2*N* differential equations for *N* biogeographic states:

$$D_{N,i}(t+\Delta t) = D_{N,i}(t) + \left[-\left(\sum_j \sum_k \lambda_{ijk} + \sum_{j \neq i} Q_{ij} + \mu_i \right) D_{N,i}(t) + \sum_{j \neq i} Q_{ij} D_{N,j}(t) + \sum_j \sum_k \lambda_{ijk} \left(D_{N,k}(t) E_j(t) + D_{N,j}(t) E_k(t) \right) \right] \Delta t$$

And,

ClaSSE models have:

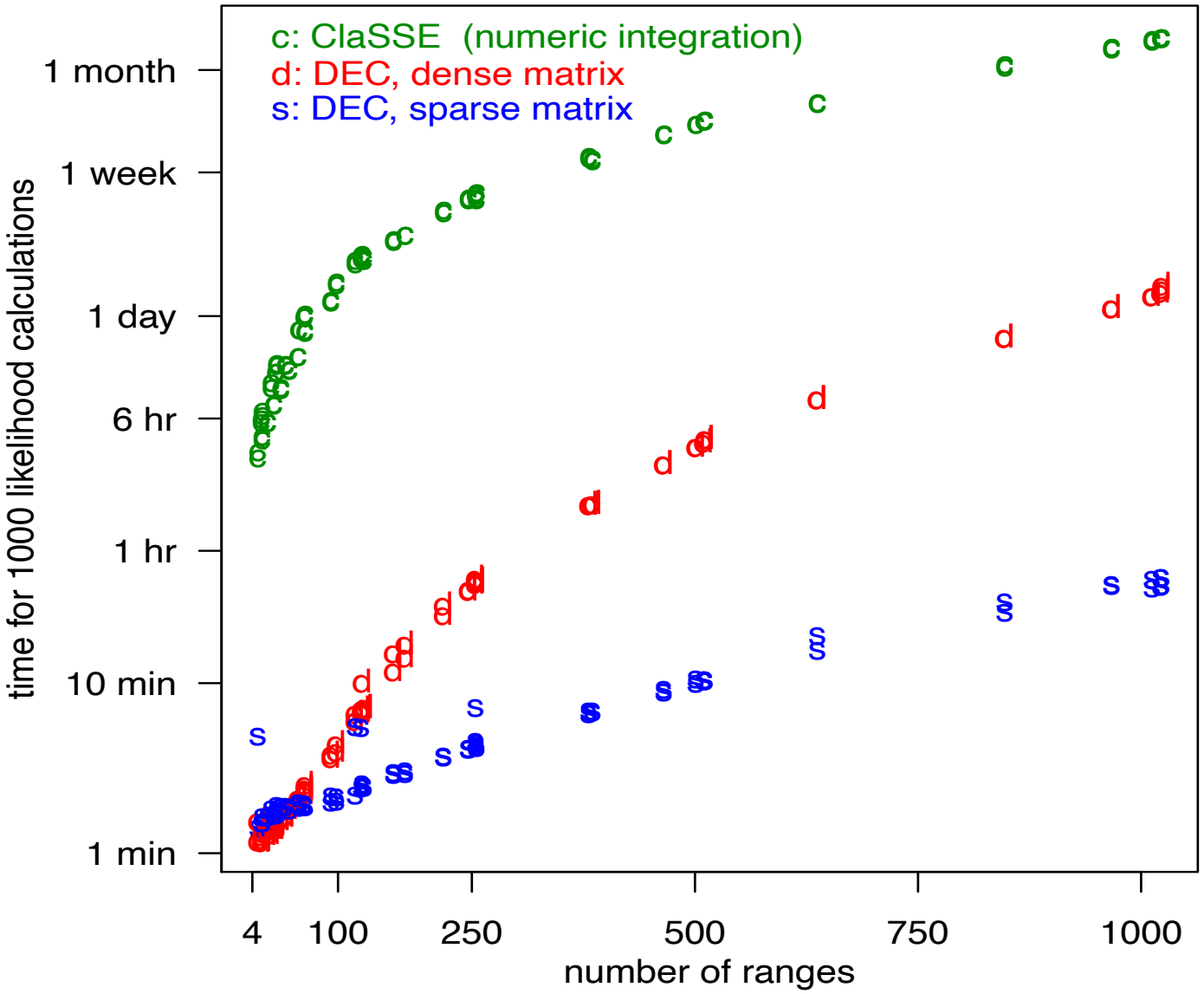
N extinction rates

N^2-*N* anagenetic transition rates

N^3 cladogenetic transition rates

State-dependent speciation-extinction models are slooooow

My first attempt – in R, but making full use of sparsity etc. to reduce ClaSSE calculations, and using R's *deSolve* library



Julia to the rescue?

Julia 1.0 released in early 2019, after many years <1.0

Core developers at MIT

Aims to have R/Python-like usage

But compiles to C-like speeds

KEY advantage: DifferentialEquations.jl library

TOOLBOX · 30 JULY 2019

Julia: come for the syntax, stay for the speed

Researchers often find themselves coding algorithms in one programming language, only to have to rewrite them in a faster one. An up-and-coming language could be the answer.

Jeffrey M. Perkel

Julia to the rescue?

KEY advantage: *DifferentialEquations.jl* library by Chris Rackauckas

https://github.com/JuliaDiffEq/DifferentialEquations.jl

Comparison Of Differential Equation Solver Software

Subject/Item	MATLAB	SciPy	deSolve	DifferentialEquations.jl	Sundials	Hairer	ODEPACK/Netlib /NAG	JITCODE	PyDSTool	FATODE	GSL	BOOST	Mathematica	Maple
Language	MATLAB	Python	R	Julia	C++ and Fortran	Fortran	Fortran	Python	Python	Fortran	C	C++	Mathematica	Maple
Selection of Methods for ODEs	Fair	Poor	Fair	Excellent	Good	Good	Good	Poor	Poor	Good	Poor	Fair	Fair	Fair
Efficiency*	Poor	Poor	Poor	Excellent	Excellent	Good	Good	Good	Good	Good	Fair	Fair	Fair	Good
Tweakability	Fair	Poor	Good	Excellent	Excellent	Good	Good	Fair	Fair	Fair	Fair	Fair	Good	Fair
Event Handling	Good	Good	Fair	Excellent	Good**	None	Good**	None	Fair	None	None	None	Good	Good
Symbolic Calculation of Jacobians and Autodifferentiation	None	None	None	Excellent	None	None	None	None	None	None	None	None	Excellent	Excellent
Complex Numbers	Excellent	Good	Fair	Good	None	None	None	None	None	None	None	Good	Excellent	Excellent
Arbitrary Precision Numbers	None	None	None	Excellent	None	None	None	None	None	None	None	Excellent	Excellent	Excellent
Control Over Linear/Nonlinear Solvers	None	Poor	None	Excellent	Excellent	Good	Depends on the solver	None	None	None	None	None	Fair	None
Built-In Parallelism	None	None	None	Excellent	Excellent	None	None	None	None	None	None	Fair	None	None
Differential-Algebraic Equation (DAE) Solvers	Good	None	Good	Excellent	Good	Excellent	Good	None	Fair	Good	None	None	Good	Good
Implicitly-Defined DAE Solvers	Good	None	Excellent	Fair	Excellent	None	Excellent	None	None	None	None	None	Good	None
Constant-Lag Delay Differential Equation (DDE) Solvers	Fair	None	Poor	Excellent	None	Good	Fair (via DDVERK)	Fair	None	None	None	None	Good	Excellent
State-Dependent DDE Solvers	Poor	None	Poor	Excellent	None	Excellent	Good	None	None	None	None	None	None	Excellent
Stochastic Differential Equation (SDE) Solvers	Poor	None	None	Excellent	None	None	None	Good	None	None	None	None	Fair	Poor
Specialized Methods for 2nd Order ODEs and Hamiltonians (and Symplectic Integrators)	None	None	None	Excellent	None	Good	None	None	None	None	None	Fair	Good	None
Boundary Value Problem (BVP) Solvers	Good	Fair	None	Good	None	None	Good	None	None	None	None	None	Good	Fair
GPU Compatibility	None	None	None	Excellent	Good	None	None	None	None	None	None	Good	None	None
Analysis Addons (Sensitivity Analysis, Parameter Estimation, etc.)	None	None	None	Excellent	Excellent	None	Good (for some methods like DASPK)	None	Poor	Good	None	None	Excellent	None

*Efficiency takes into account not only the efficiency of the implementation, but the features of the implemented methods (advanced timestepping controls, existence of methods which are known to be more efficient, Jacobian handling)

**Event handling needs to be implemented yourself using basic rootfinding functionality

Scale	None	Poor	Fair	Good	Excellent
Explanation	Functionality does not exist	Functionality exists, but is feature-incomplete	The basic features exist	The basic features exist and some extra tweakability exists. May include extra methods for efficiency.	Has all of the basic features and more. Extra features for flexibility and efficiency.

For more detailed explainations and comparisons, see the following blog post:

http://www.stochasticlifestyle.com/a-comparison-between-differential-equation-solver-suites-in-matlab-r-julia-python-c-and-fortran

https://github.com/JuliaDiffEq/DifferentialEquations.jl

MIT News: New climate model to be built from the ground up

December 12, 2018

Our group collaborates in a new Climate Modeling Alliance to advance climate modeling and prediction. You can read about the overall project in a press release issued by Caltech, in partnership with the MIT School of Science, the Naval Postgraduate School, and the Jet Propulsion Laboratory. We also have a brand new website with more details.

The MIT team's role in CliMA is described in an article published on the PAOC's website and reproduced here:

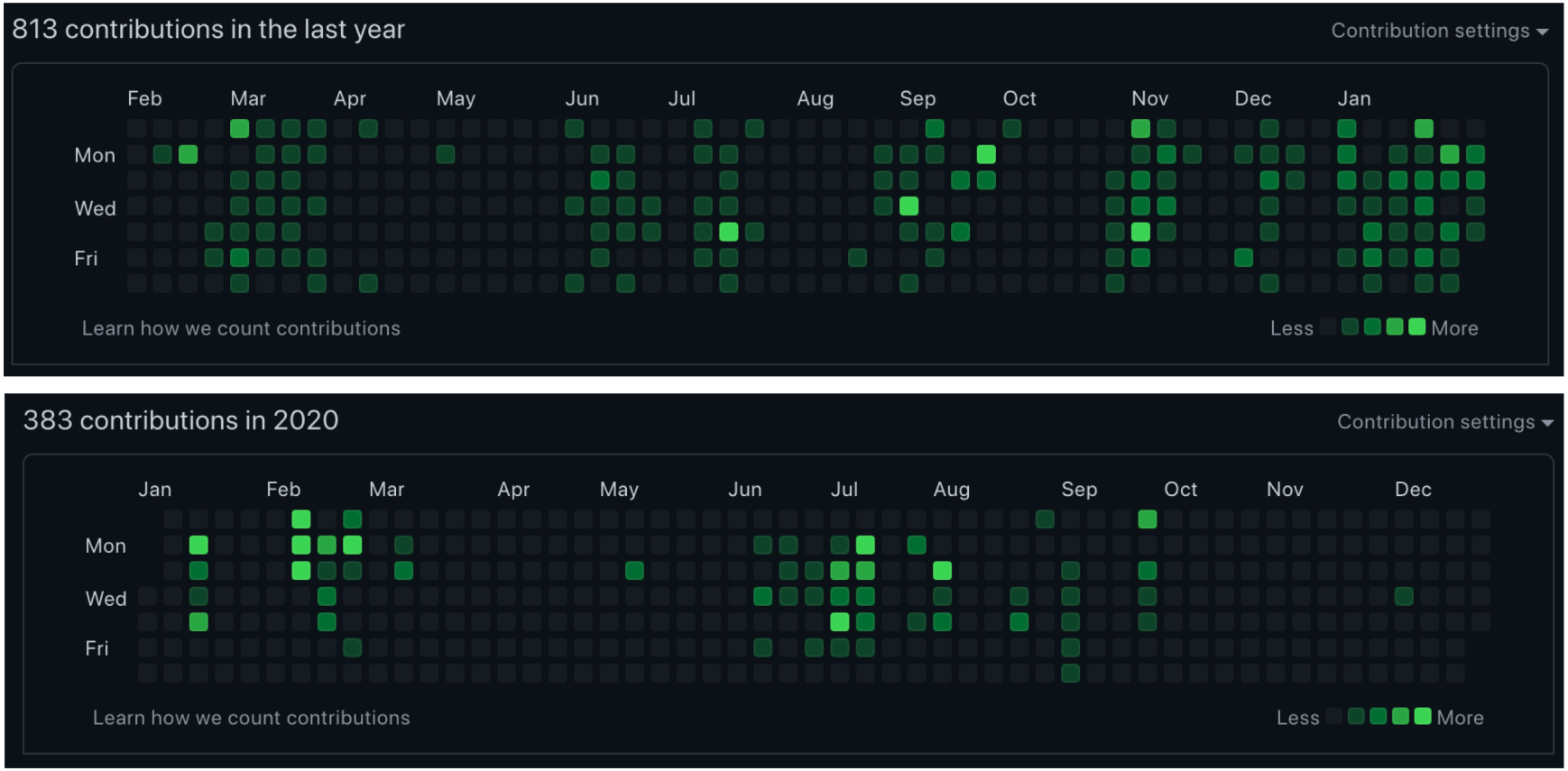
At MIT, Professors Raffaele Ferrari and John Marshall, both Cecil and Ida Green Professors of Oceanography and members of the Program in Atmospheres, Oceans and Climate (PAOC), will develop the ocean and sea-ice component of the CliMa climate model, leveraging their expertise in the development of ocean models (the MITgcm was spearheaded by Marshall, with Chris Hill and Jean Michel Campin as key developers) and in the representation of sub-gridscale mixing processes in the ocean (Ferrari and Marshall).

To take advantage of new computer architectures, languages, and machine learning techniques, the team has partnered with Alan Edelman's group in CSAIL at MIT, who will help write the new generation climate model in the Julia computing language developed by the group. This will enable the MIT team to target GPUs, CPUs, and evolving computer architectures within one code

ıs.jl

What did all of this *PhyBEARS* Julia coding do?

GitHub commits to *PhyBEARS.jl*



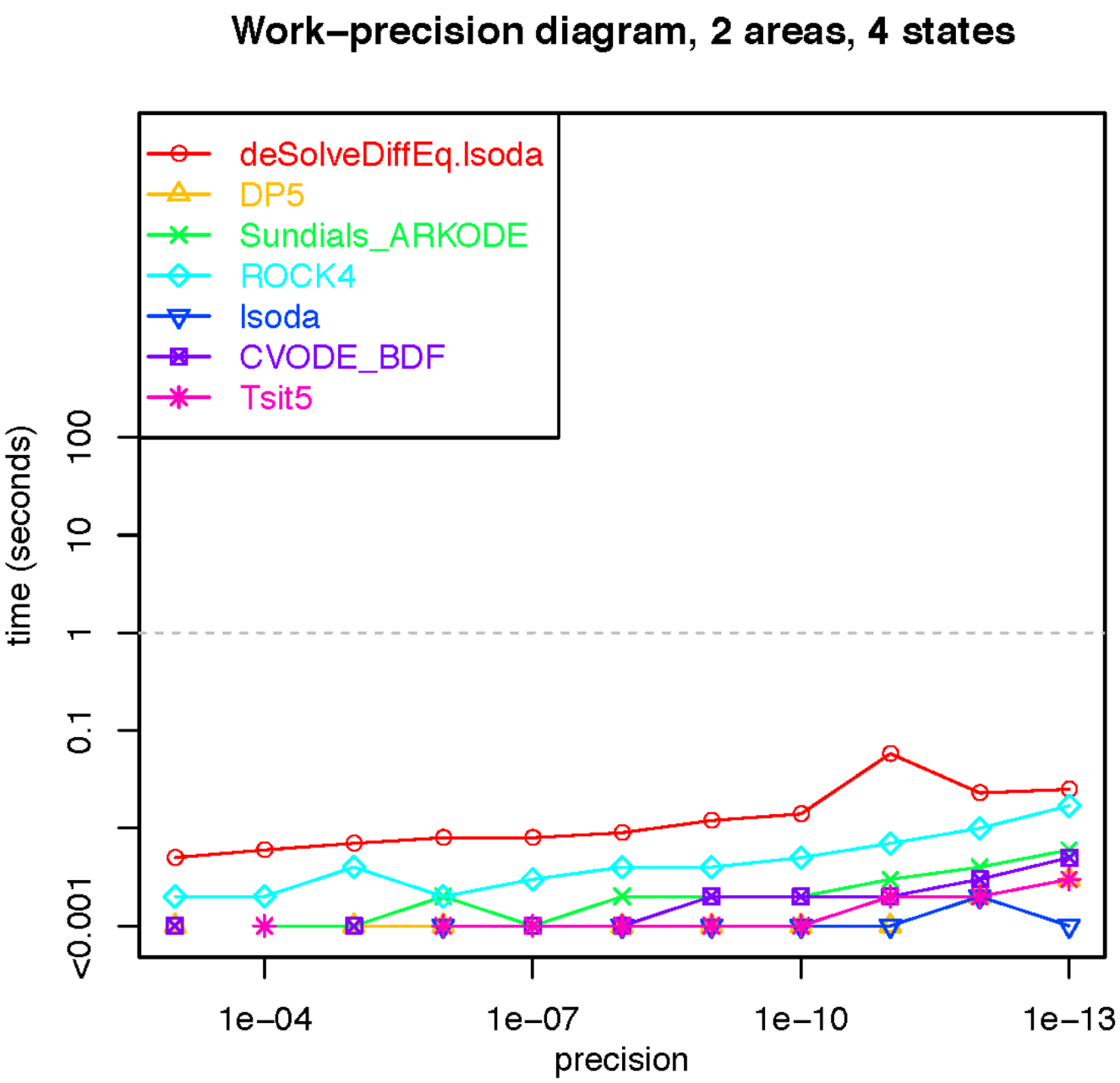
https://github.com/nmatzke

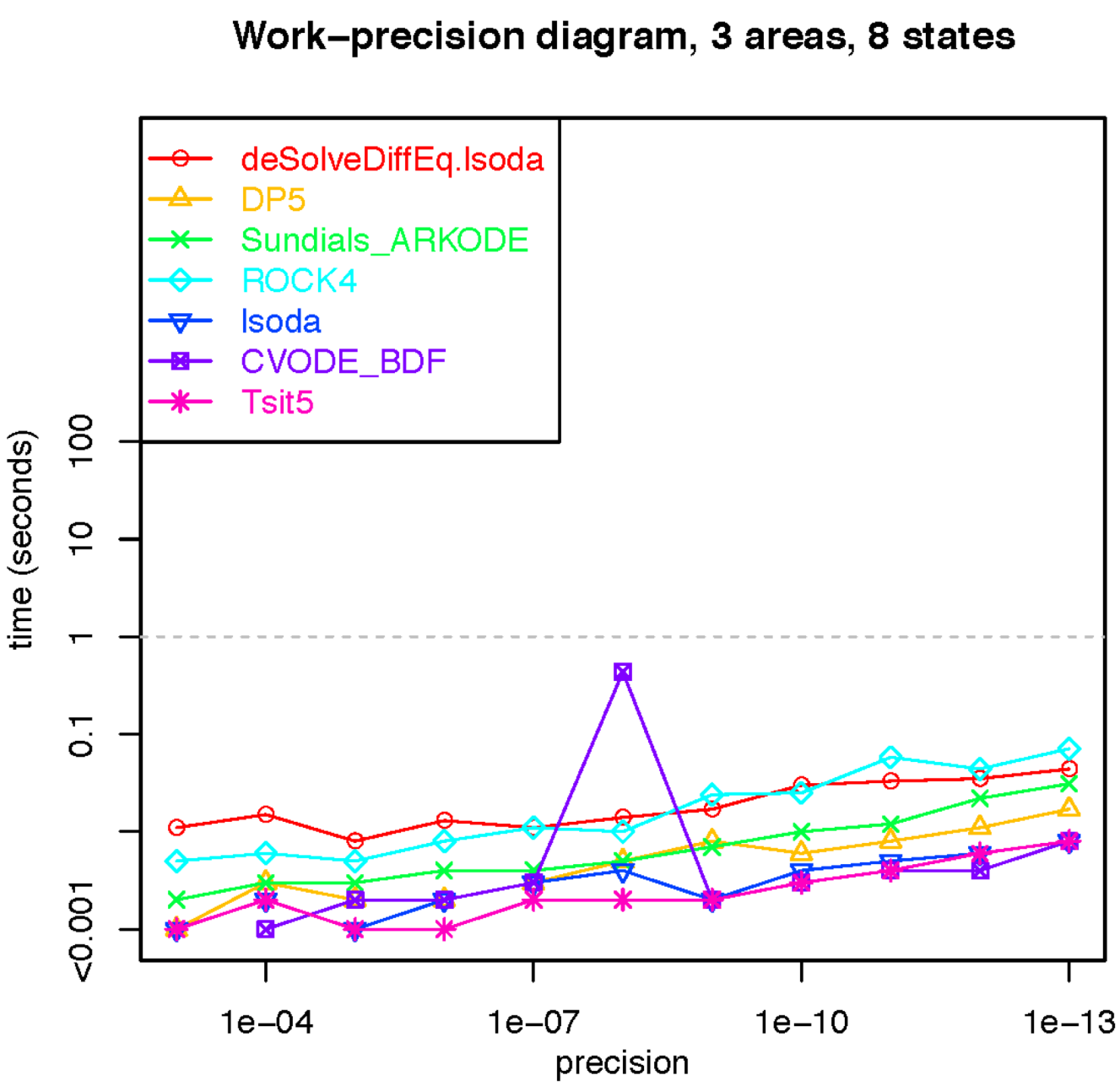
ClaSSE model in *Julia*

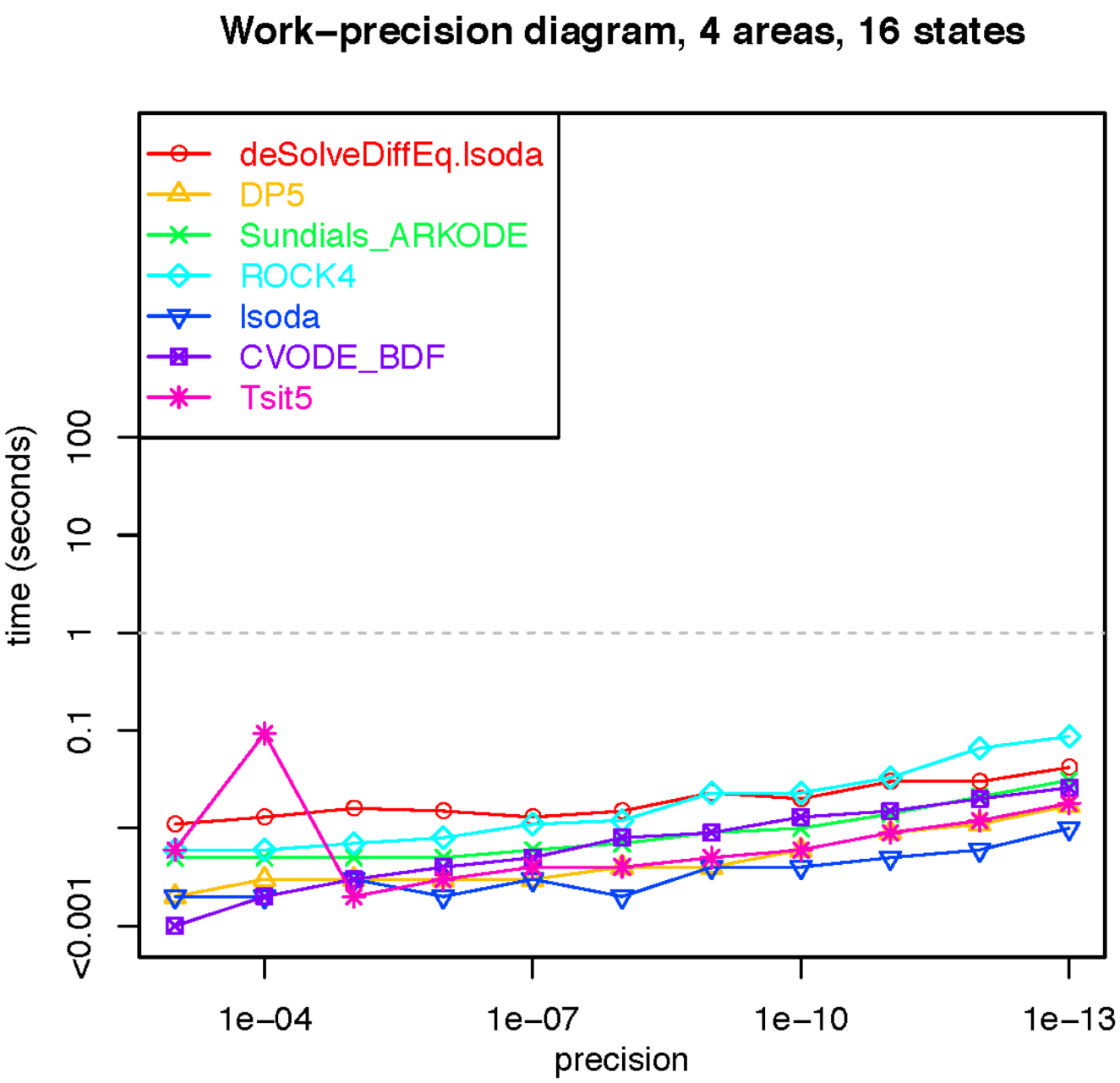
Methods:

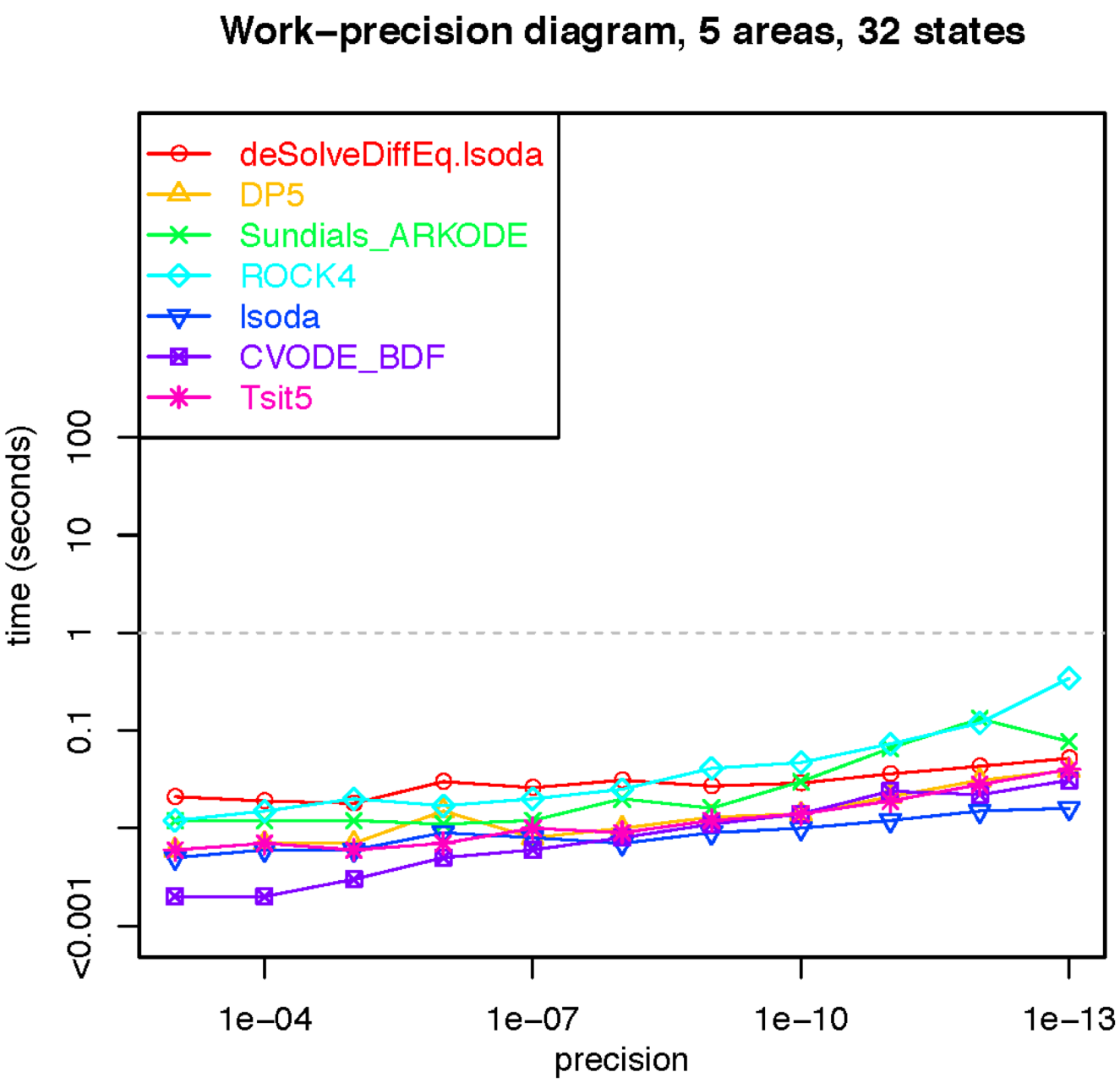
1. Compare the many solvers available in Julia's DifferentialEquations.jl library

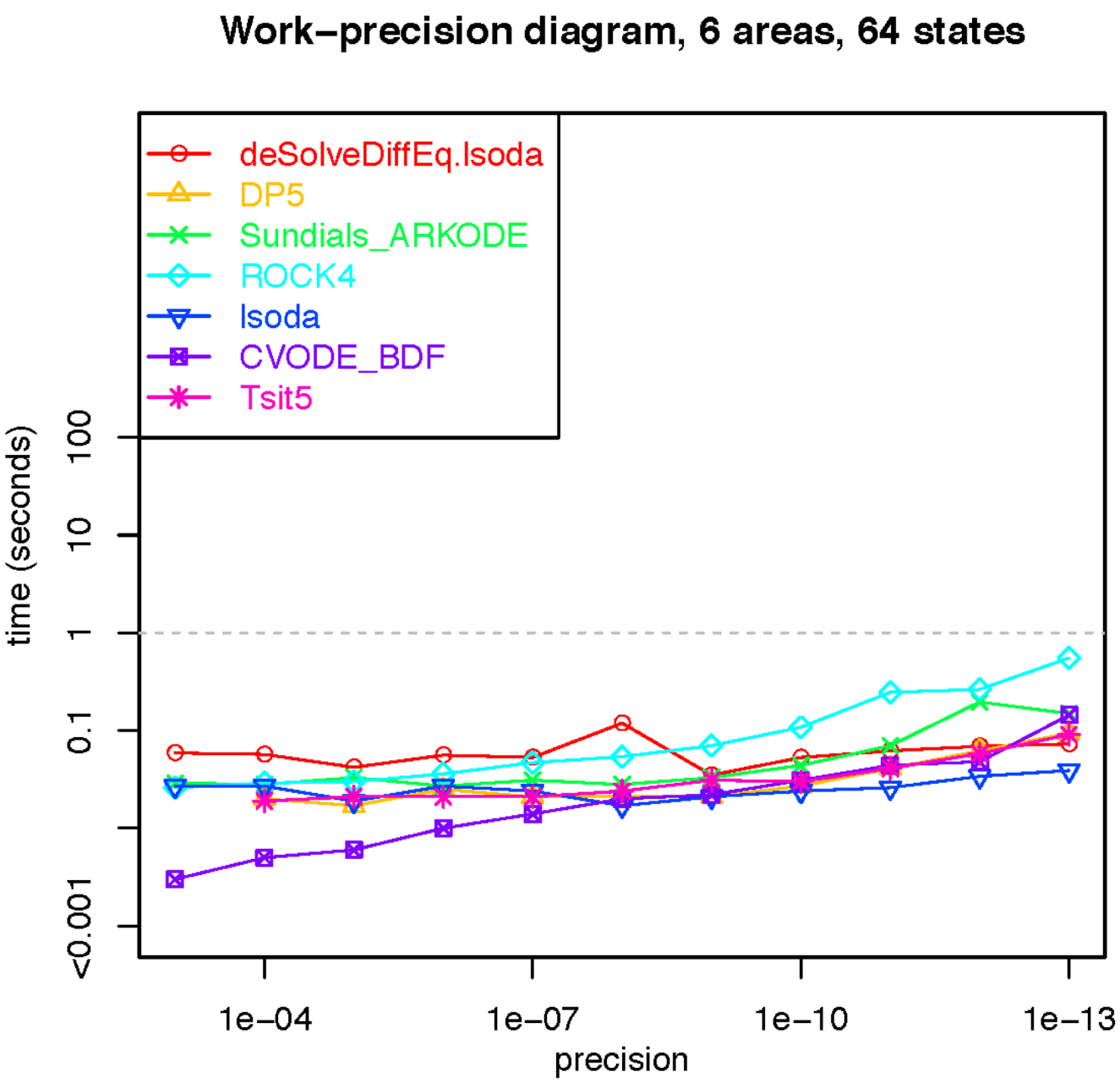
https://github.com/JuliaDiffEq/DifferentialEquations.jl

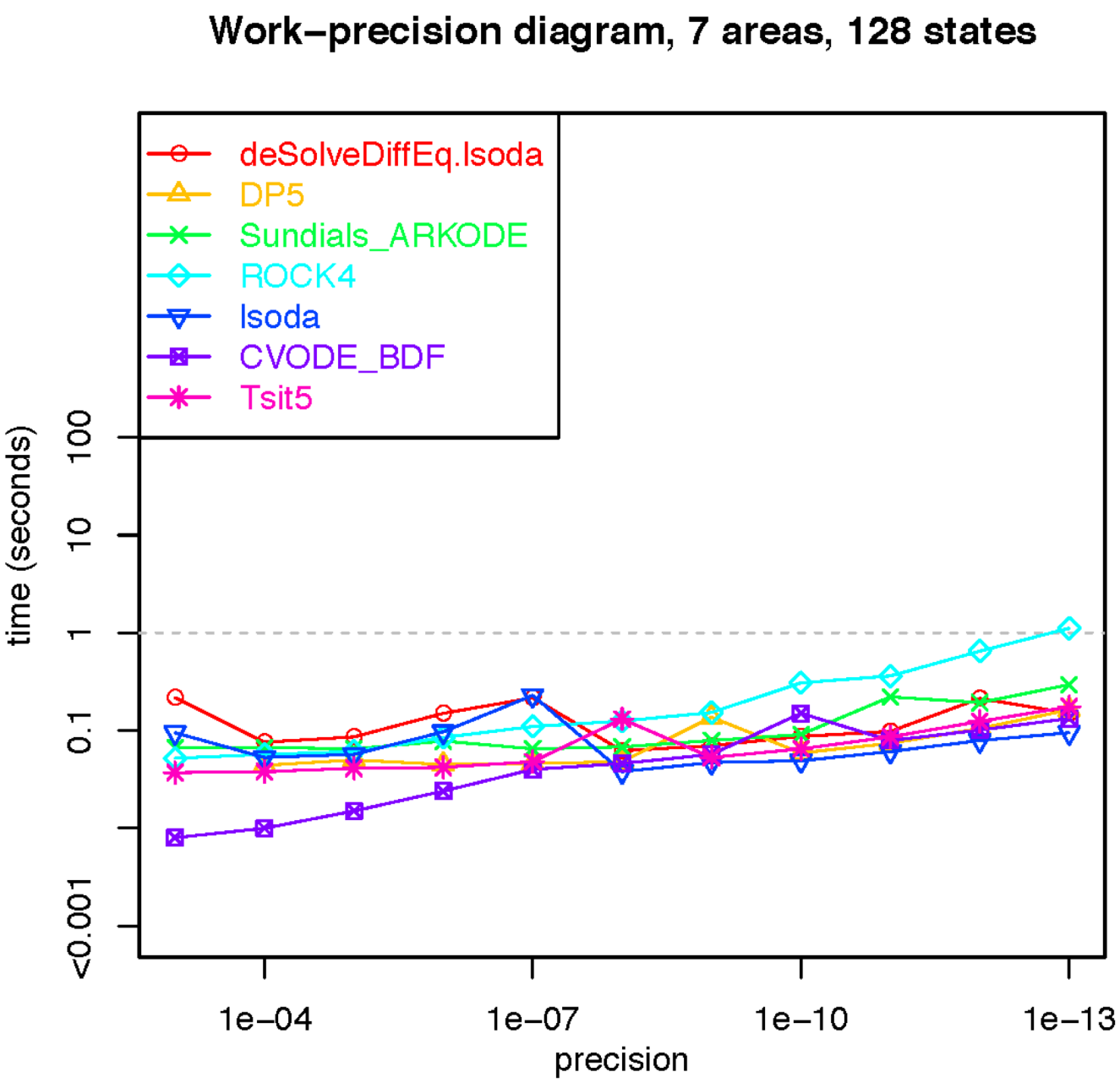

Work−precision diagram, 2 areas, 4 states
deSolveDiffEq.lsoda
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
<0.001
0.1
1
10
100
1e−04
1e−07
1e−10
1e−13
precision

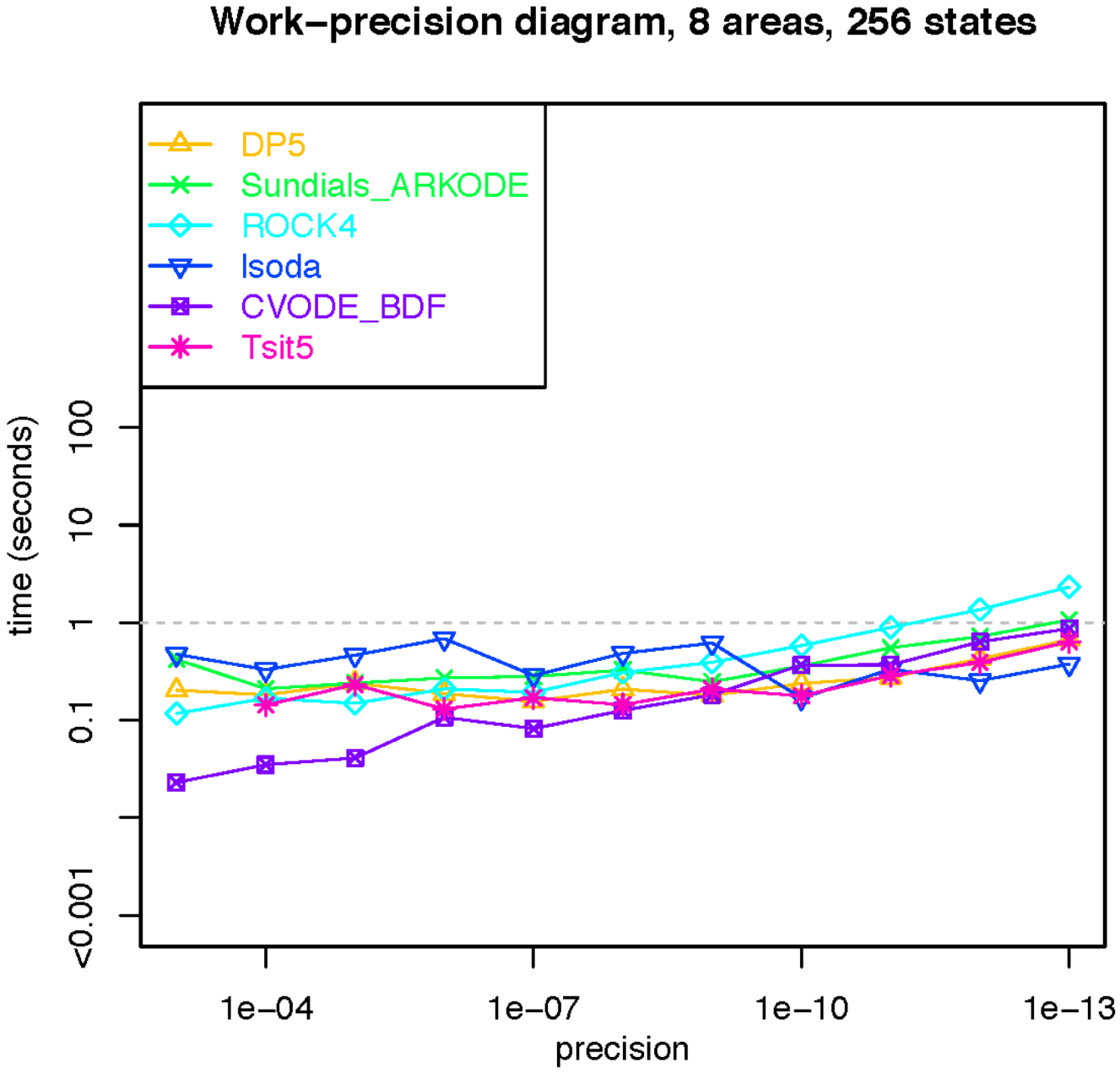

Work–precision diagram, 3 areas, 8 states
deSolveDiffEq.lsoda
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
100
10
1
0.1
<0.001
1e−04
1e−07
1e−10
1e−13
precision

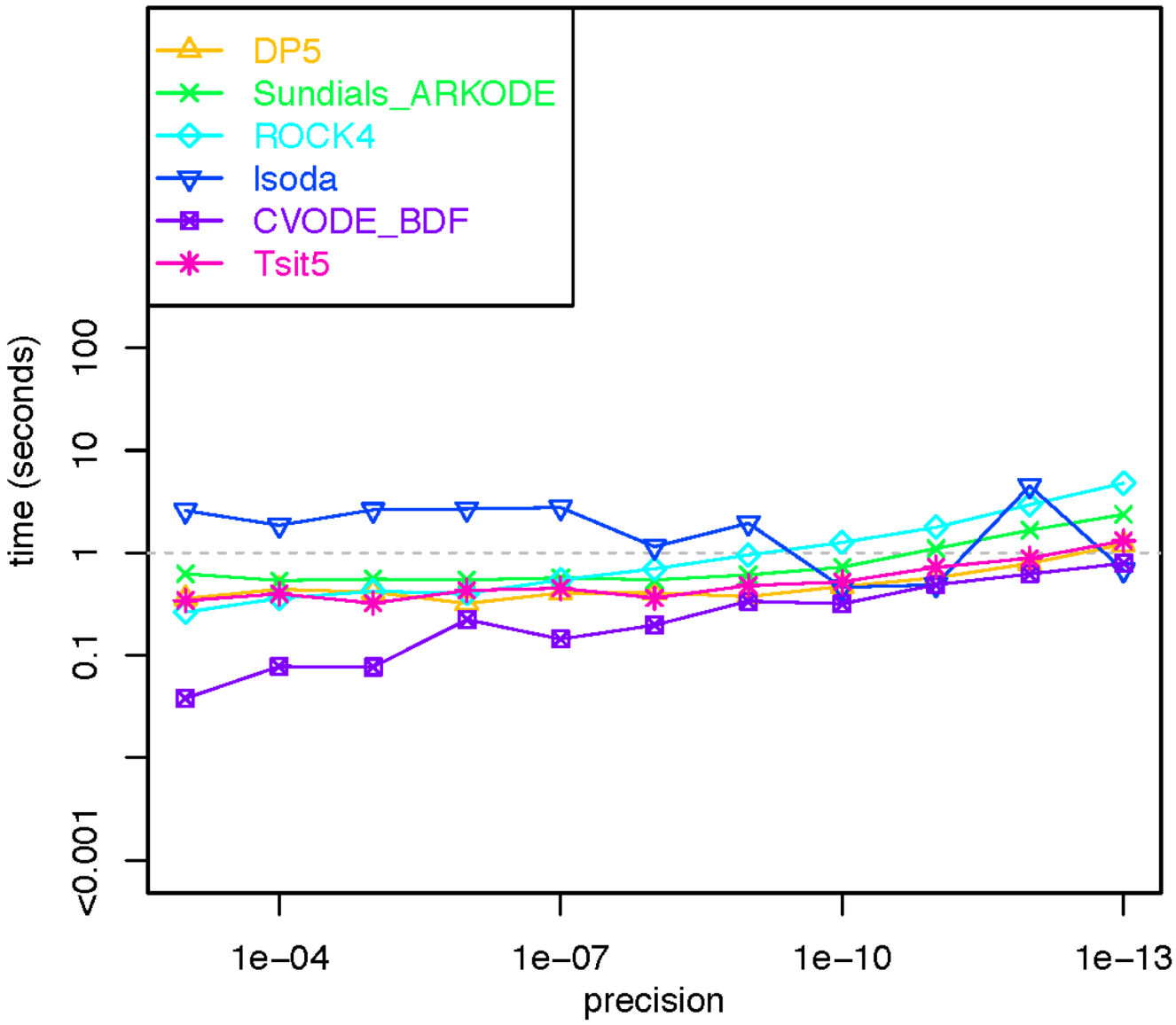

Work−precision diagram, 4 areas, 16 states
deSolveDiffEq.lsoda
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
100
10
1
0.1
<0.001
1e−04
1e−07
1e−10
1e−13
precision

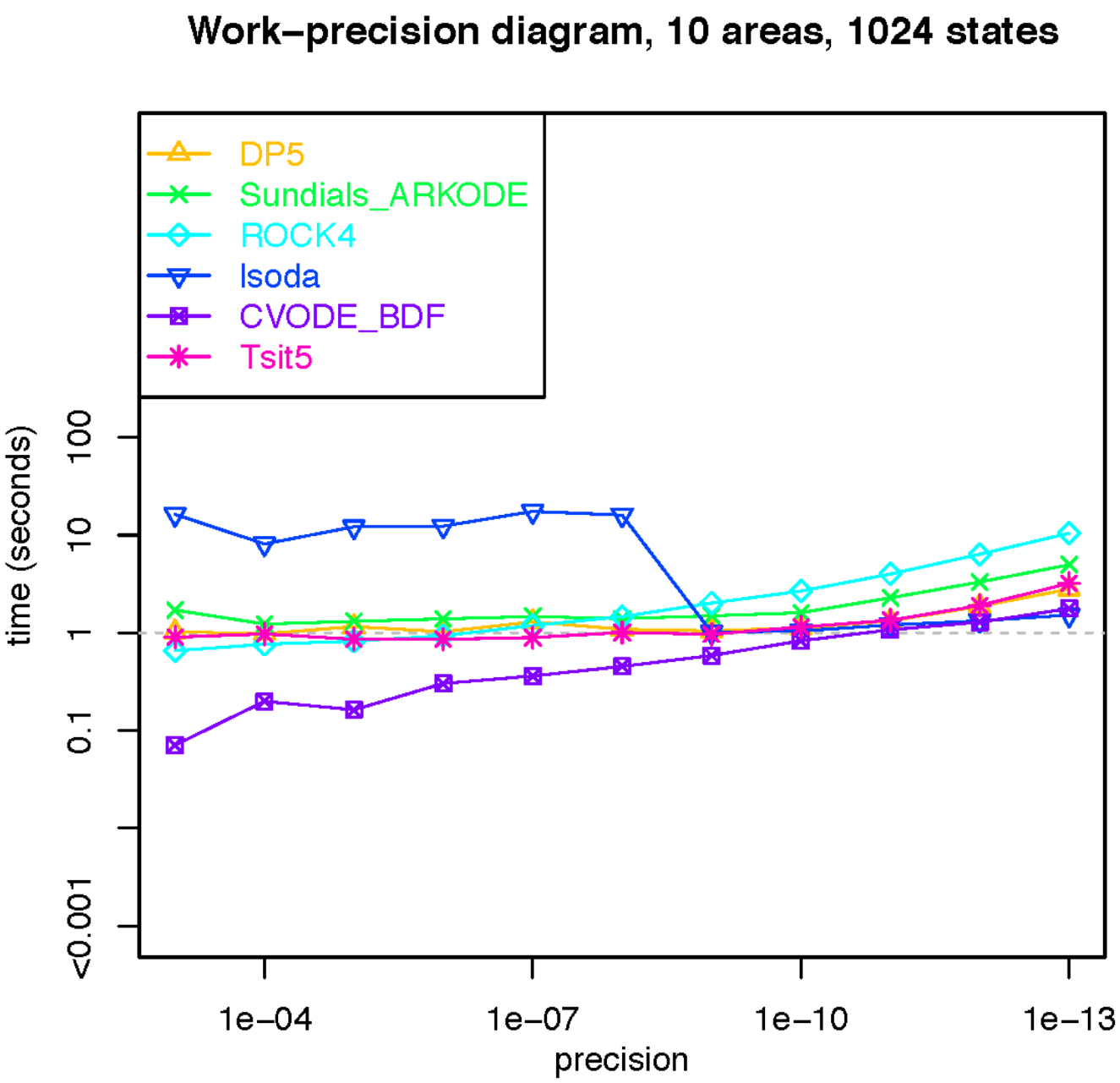

Work−precision diagram, 5 areas, 32 states
deSolveDiffEq.lsoda
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
100
10
1
0.1
<0.001
1e−04
1e−07
1e−10
1e−13
precision

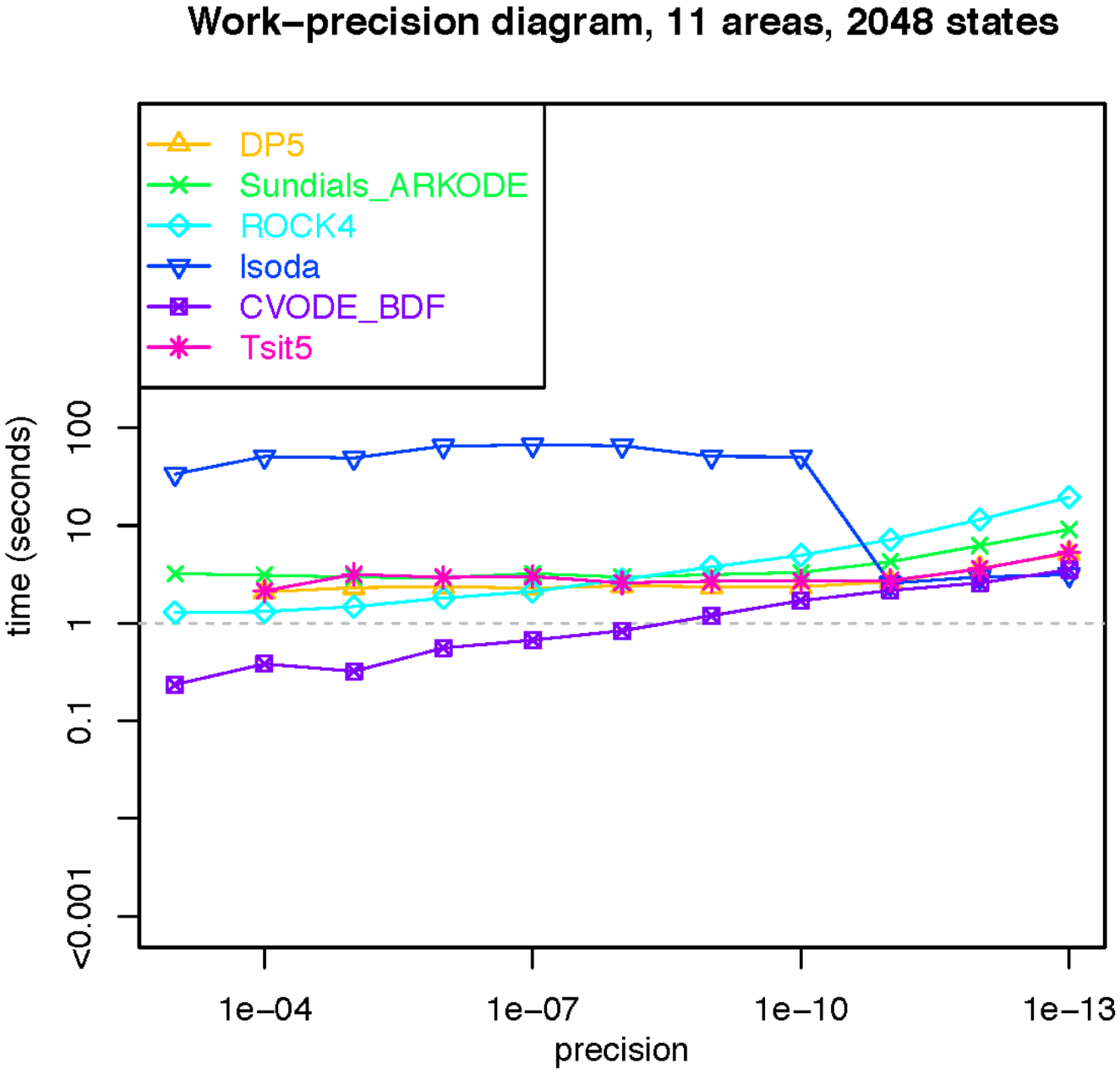

Work−precision diagram, 6 areas, 64 states
deSolveDiffEq.lsoda
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
100
10
1
0.1
<0.001
1e−04
1e−07
1e−10
1e−13
precision


Work−precision diagram, 7 areas, 128 states
deSolveDiffEq.lsoda
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
100
10
1
0.1
<0.001
1e−04
1e−07
1e−10
1e−13
precision


Work−precision diagram, 8 areas, 256 states
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
100
10
1
0.1
<0.001
1e−04
1e−07
1e−10
1e−13
precision


Work−precision diagram, 9 areas, 512 states
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
<0.001
0.1
1
10
100
1e−04
1e−07
1e−10
1e−13
precision


Work–precision diagram, 10 areas, 1024 states
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
<0.001
0.1
1
10
100
1e−04
1e−07
1e−10
1e−13
precision


Work–precision diagram, 11 areas, 2048 states
DP5
Sundials_ARKODE
ROCK4
lsoda
CVODE_BDF
Tsit5
time (seconds)
<0.001
0.1
1
10
100
1e−04
1e−07
1e−10
1e−13
precision

ClaSSE model in *Julia*

Methods:

3. Made basic calculation as fast as possible in Julia
   - fully used sparsity
   - all parameter arrays & indexing pre-allocated
   - minimized Garbage Collection with @benchmark & advice from DifferentialEquations help forum (including Rackauckas)
4. Parallelized calculation

https://github.com/JuliaDiffEq/DifferentialEquations.jl

ClaSSE model in *Julia*

Methods:

5. Precalculate “Es” (conditional extinction probabilities) once, then interpolate for “Ds” (observed data probabilities)
6. Economize inner loops (20x speed improvement)

7. Implement “G Flow” precalculation

https://github.com/JuliaDiffEq/DifferentialEquations.jl

Added Louca-Pennell method

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DOI:10.1093/sysbio/syz055

A General and Efficient Algorithm for the Likelihood of Diversification and Discrete-Trait Evolutionary Models

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Associate Editor: Stephen Smith

Abstract.—As the size of phylogenetic trees and comparative data continue to grow
developed to investigate the processes that gave rise to them, macroevolutionary ana
limited by computational requirements. Here, we introduce a novel algorithm, based
equations that describe likelihoods along tree edges in backward time, to reduce redunda
compute the likelihood of various macroevolutionary models. Our algorithm applies
including birth–death models and models that account for state- or time-dependent rates
models of discrete-trait evolution, and provides an alternative way to describe macroevo
demonstration of our algorithm's utility, we implemented it for a popular class of state-de
BiSSE, MuSSE, and their extensions to hidden-states. Our implementation is available th
show that, for these models, our algorithm is one or more orders of magnitude faster tha
applied to large phylogenies. Our algorithm thus enables the fitting of state-dependent
massive phylogenies with millions of tips and may lead to potentially similar computatio
macroevolutionary models. [Dynamical systems theory; extinction; flow; likelihood; mac

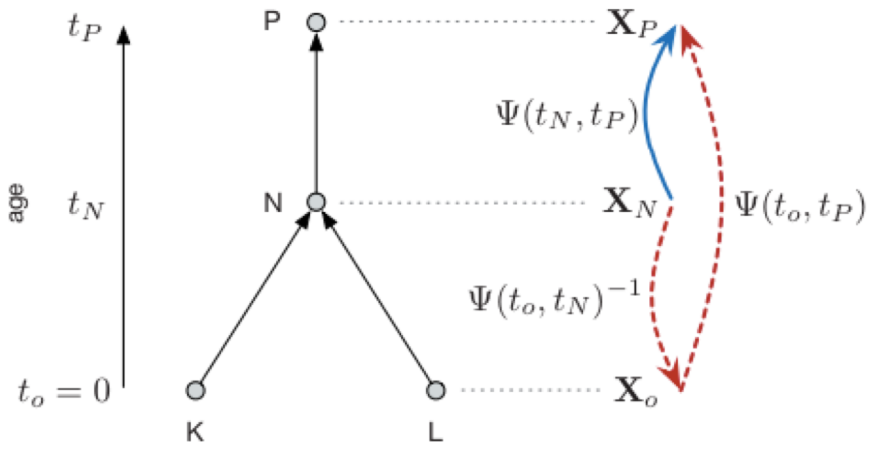
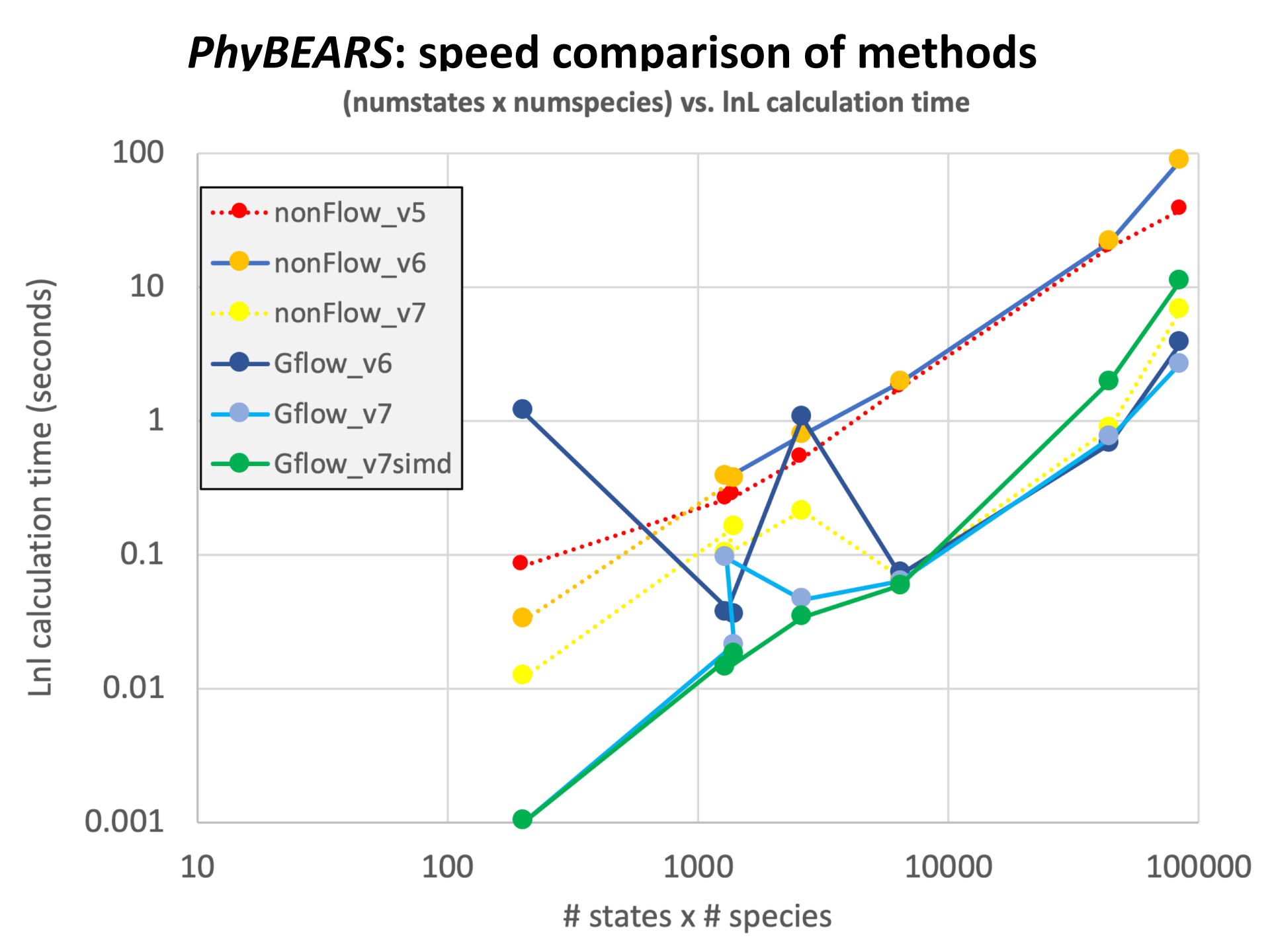


FIGURE 1. Conceptual illustration of the revised algorithm. Our proposed algorithm can be used to rapidly compute variables (e.g., state likelihoods) at nodes that are *a priori* defined as solutions of a linear differential equation along edges. In the example illustrated, as in dSSE models, the initial condition for the vector-valued variable $\mathbf{X}$ at node N is determined based on the values of $\mathbf{X}$ calculated previously on the descending edges (connecting tips L and K). The goal is to obtain the solution $\mathbf{X}_P$ of the differential equation along edge $N \rightarrow P$, given the initial condition $\mathbf{X}_N$ at node N. Instead of explicitly solving this differential equation along the edge, one can calculate $\mathbf{X}_P$ as $\mathbf{X}_P = \Psi(t_o, t_P) \cdot \Psi(t_o, t_N)^{-1} \cdot \mathbf{X}_N$, where t_o is some starting time point (e.g., $t_o = 0$) for which the flow $\Psi(t_o, t)$ has been previously calculated. The direct mapping $\Psi(t_N, t_P)$ (continuous arrow) is equivalent to the product $\Psi(t_o, t_P) \cdot \Psi(t_o, t_N)^{-1}$ (dashed arrows), and hence an explicit calculation of $\Psi(t_N, t_P)$ is not necessary.


PhyBEARS: speed comparison of methods
(numstates x numspecies) vs. lnL calculation time
nonFlow_v5
nonFlow_v6
nonFlow_v7
Gflow_v6
Gflow_v7
Gflow_v7simd
lnl calculation time (seconds)
100
10
1
0.1
0.01
0.001
10
100
1000
10000
100000
states x # species

PhyBEARS: Likelihoods match across methods as well!

Also checked against:

BioGeoBEARS
BiSSE (diversitree)
ClaSSE (diversitree)

nonFlow_v5	-115.99314
nonFlow_v6	-115.99314
nonFlow_v7	-115.99314
Gflow_v6	-115.99315
Gflow_v7	-115.99315
Gflow_v7simd	-115.99315
nonFlow_v5	-1386.5099
nonFlow_v6	-1386.5099
nonFlow_v7	-1386.5099
Gflow_v6	-1386.51
Gflow_v7	-1386.51
Gflow_v7simd	-1386.51
nonFlow_v5	-1234.4104
nonFlow_v6	-1234.4104
nonFlow_v7	-1234.4104
Gflow_v6	-1234.4105
Gflow_v7	-1234.4105
Gflow_v7simd	-1234.4105
nonFlow_v5	-2321.038
nonFlow_v6	-2321.038
nonFlow_v7	-2321.038
Gflow_v6	-2321.038
Gflow_v7	-2321.038
Gflow_v7simd	-2321.038
nonFlow_v5	-191.10675
nonFlow_v6	-191.10675
nonFlow_v7	-191.10675
Gflow_v6	-191.10674
Gflow_v7	-191.10674
Gflow_v7simd	-191.10674
nonFlow_v5	-2064.0736
nonFlow_v6	-2064.0736
nonFlow_v7	-2064.0736
Gflow_v6	-2064.0735
Gflow_v7	-2064.0735
Gflow_v7simd	-2064.0735
nonFlow_v5	-3942.9718
nonFlow_v6	-3942.9718
nonFlow_v7	-3942.9718
Gflow_v6	-3942.9716
Gflow_v7	-3942.9716
Gflow_v7simd	-3942.9716

Additional progress

Ph.D. student Wallis Bland

Integrating *Gplates* and *PhyBEARS*

Simulation-Inference tests of *PhyBEARS*

Wallis Bland

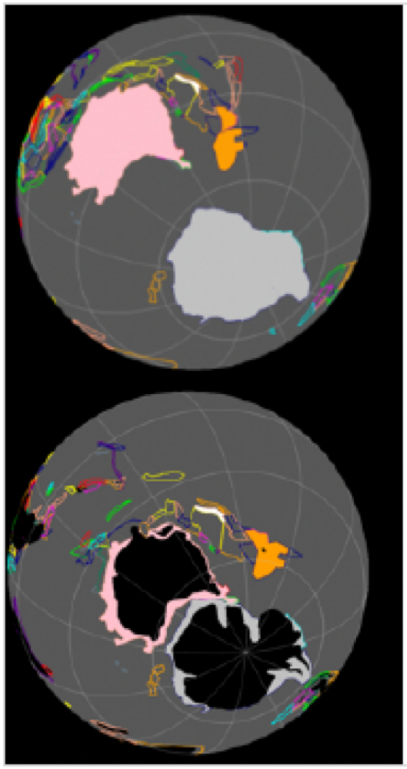
Additional progress

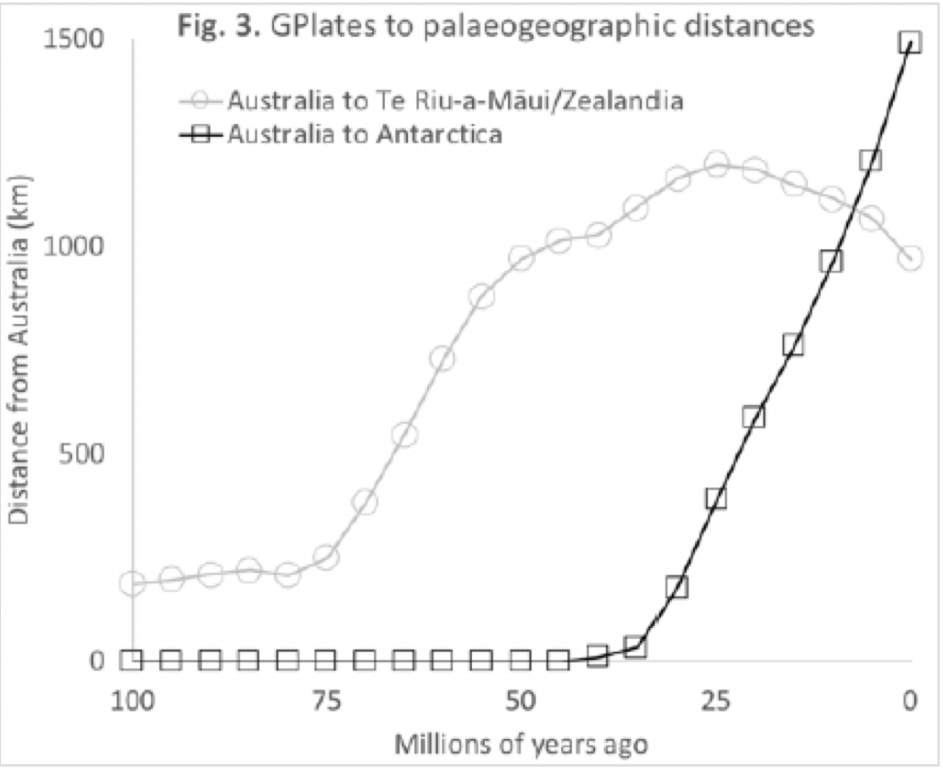
Ph.D. student Wallis Bland

Integrating *Gplates* and *PhyBEARS*

Simulation-Inference tests of *PhyBEARS*

Wallis Bland





Additional progress: Inference under changing geog.

Using Julia's amazing capabilities for including interpolators in ODEs:

Time

GPlates-> Paleodistances

Dispersal rates

Changing area sizes

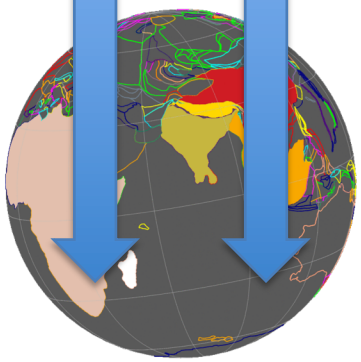
Changing extinction rates

ODEs for "Es" (extinction probs.)

ODEs for "Ds" (data probs.)

A=dG matrix

Gflow matrix



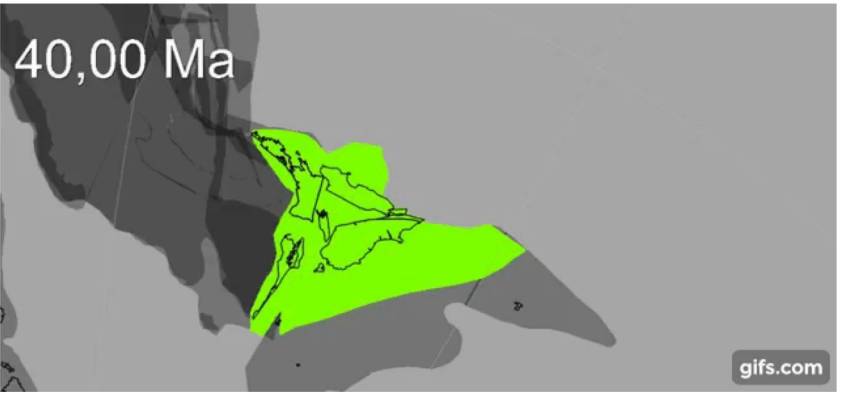
LnL: Log-likelihood

PhyBEARS: complexifying biogeography

Summary:

Integrated *PhyBEARS* with *GPlates* for continuous palaeogeography

GPlates animation: Zealandia crustal thickness



Müller, Sahirovic et al. (2018), U. Sydney "Modelling and visualising distributed crustal deformation of Australia and Zealandia using *GPlates 2.0*." https://www.youtube.com/watch?v=iKkCIdZ6oaE

PyGPlates: extract & use continuous estimates of areas, distances

Sabin Zahirovic, *GPlates* coauthor, U. Sydney

Comparison of model features

	Features of biogeographical models: / Models in use:	*Markov*-k (and e.g. DNA models) Lewis (2001)	*Lagrange* / DEC (Python) Ree & Smith (2008)
↓ Increasing realism ↓	Probabilistic model of range evolution	X	X
	Ranges can occupy multiple areas		X
	Range expansion/contraction		X
	Vicariance (large range can divide at speciation)		X
	Plate tectonics/paleogeography (by hand)		X
	Jump dispersal		
	Statistical comparison of multiple models		
	Distance-dependent dispersal		
	Trait-dependent dispersal		
	Speciation rate depends on region		
	Lineage extinction		
	Incomplete sampling in some subclades		
	Fossil sampling		
	Plate tectonics/paleogeography (using *GPlates* software)		
Speed	Sparse matrix library to speed up large matrices		
	Example number of areas	20	11
	State space (maximum number of possible ranges in a typical analysis)	20	2048
	Calculation strategy	matrix exponen	
	Analysis time	10 min.	1 week

X=

Conclusions

1. Large ClaSSE models will probably never be super-fast – but Julia versions should be fast enough for e.g. Maximum Likelihood

2. The number of free parameters explodes to meaninglessness quickly. So model (parametric) complexity has to be carefully controlled by carefully thinking about, and designing, meaningful models controlled by a small number of parameters (as in e.g. standard biogeography models)

3. Julia's DifferentialEquations solvers are *very* flexible, suggesting that more interesting models could be designed in this framework (regional carrying capacity, interpolating in the impact of external forcers like changing geography & climate, etc.)

Acknowledgements: Marsden grant no. 18-UOA-34